

SET 68 • 4.MD.A.3

BACKWARD 2

Add up to four two-digit numbers • Grade 2 • Number & Operations in Base Ten
• 2.NBT.B.6

Add.

$$12 + 5 + 12 + 5$$

Sum: _____

True or false?

$$25 + 40 + 25 + 40 = 130$$

A. True

B. False

$$9 + 9 + 3 + 3 = ?$$

A. 24

B. 12

C. 21

D. 81

Add.

$$15 + 8 + 15 + 8$$

Sum: _____

True or false?

$$14 + 6 + 14 + 6 = 30$$

A. True

B. False

SET 68 • 4.MD.A.3

FORWARD 1

Find the perimeter of shapes with fractional side lengths • Grade 5 • Number & Operations–Fractions • 5.NF.A.2

A rectangle is $\frac{3}{4}$ m long and $\frac{1}{2}$ m wide.

What is its perimeter?

Perimeter: _____

True or false?

A square with sides of $\frac{2}{3}$ foot has a perimeter of $2\frac{2}{3}$ feet.

- A. True
- B. False

A rectangular card is $2\frac{1}{2}$ inches long and $1\frac{1}{4}$ inches wide.

What is its perimeter?

A. $3\frac{3}{4}$ inches

B. $3\frac{1}{8}$ inches

C. $6\frac{1}{2}$ inches

D. $7\frac{1}{2}$ inches

A triangle has sides of $\frac{1}{2}$ yd, $\frac{2}{3}$ yd, and $\frac{5}{6}$ yd.

What is its perimeter?

Perimeter: _____

True or false?

A rectangle that is $\frac{1}{3}$ m by $\frac{1}{4}$ m has a perimeter of $\frac{7}{12}$ m.

A. True

B. False

SET 68 • 4.MD.A.3

FORWARD 2

Evaluate perimeter formulas for given side lengths • Grade 6 • Expressions & Equations • 6.EE.A.2.c

Use $P = 2l + 2w$ to find the perimeter when $l = 8.5$ cm and $w = 4$ cm.

Perimeter: _____

True or false?

For a square, $P = 4s$. When $s = 2.25$ m, $P = 9$ m.

- A. True
- B. False

Use $P = 2(l + w)$.

What is P when $l = 12$ and $w = 7.5$?

A. 19.5

B. 90

C. 39

D. 29

Evaluate $2l + 2w$ when $l = \frac{3}{4}$ and $w = \frac{1}{2}$.

Value: _____

True or false?

When $l = 10$ and $w = 3$, $2l + 2w = 23$.

A. True

B. False

Find an unknown side length from the area of a rectangle

MAIN	Grade 4 main question	4.MD.A.3
BACKWARD 2	Find the unknown factor	3.OA.A.4
BACKWARD 3	Find the area of a rectangle by multiplying its side lengths	3.MD.C.7.b
FORWARD 1	Find a missing dimension when the volume is known	5.MD.C.5.b
FORWARD 2	Solve equations of the form $px = q$ for a missing measurement	6.EE.B.7

SET 49 • 4.MD.A.3

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle has an area of 72 square feet and a length of 9 feet.

What is its width?

Width: _____

A rectangle has an area of 63 square cm and a length of 9 cm.

What is its width?

Width: _____

A rectangular patio has an area of 96 square meters.
One side is 12 m long.

How long is the other side?

A. 36 m

B. 108 m

C. 84 m

D. 8 m

True or false?

A square has an area of 36 square inches, so each side is 6 inches long.

A. True

B. False

A rectangle has an area of 54 square units and a width of 6 units.

Write an equation with a letter for the length. Then solve it.

Equation: _____

Length: _____

SET 49 • 4.MD.A.3

BACKWARD 2

Find the unknown factor • Grade 3 • Operations & Algebraic Thinking •
3.OA.A.4

Find the unknown number.

$$8 \times ? = 64$$

Unknown: _____

True or false?

If $9 \times w = 45$, then $w = 5$.

A. True

B. False

$$? \times 7 = 56$$

A. 7

B. 8

C. 49

D. 63

Find the unknown number.

$$6 \times ? = 42$$

Unknown: _____

True or false?

If $4 \times n = 28$, then $n = 6$.

A. True

B. False

SET 49 • 4.MD.A.3

BACKWARD 3

Find the area of a rectangle by multiplying its side lengths • Grade 3 •
Measurement & Data • 3.MD.C.7.b

A rectangle is 8 cm long and 7 cm wide.

What is its area?

Area: _____

True or false?

A 9-by-6 rectangle has an area of 54 square units.

A. True

B. False

A rug is 5 feet by 8 feet.

What is its area?

- A.** 45 square feet
- B.** 40 square feet
- C.** 13 square feet
- D.** 26 square feet

A garden is 9 m long and 4 m wide.

What is its area?

Area: _____

True or false?

A 7-by-7 square has an area of 14 square units.

A. True

B. False

SET 49 • 4.MD.A.3

FORWARD 1

Find a missing dimension when the volume is known • Grade 5 • Measurement & Data • 5.MD.C.5.b

A box has a volume of 60 cubic inches. It is 5 in. long and 3 in. wide.

How tall is it?

Height: _____

True or false?

A prism with a volume of 48 cubic cm and a base area of 12 square cm is 4 cm tall.

- A. True
- B. False

A tank holds 120 cubic feet. Its base is 6 ft by 5 ft.

How deep is the tank?

- A.** 20 feet
- B.** 4 feet
- C.** 10 feet
- D.** 109 feet

A prism has a volume of 90 cubic meters and a height of 5 m.

What is the area of its base?

Base area: _____

True or false?

A box that is 4 units by 3 units at the base and has a volume of 36 cubic units is 4 units tall.

- A. True
- B. False

SET 49 • 4.MD.A.3

FORWARD 2

Solve equations of the form $px = q$ for a missing measurement • Grade 6 •
Expressions & Equations • 6.EE.B.7

A rectangle has an area of 7.5 square meters and a length of 3 m.

Write and solve an equation for its width w .

Equation: _____

$W =$ _____

True or false?

A triangle has an area of 24 square units and a base of 8 units. Solving $4h = 24$ gives a height of 6 units.

- A.** True
- B.** False

Solve for b.

$$12b = 84$$

A. 1,008

B. 7

C. 72

D. 96

Solve for x.

$$\frac{1}{2}x = 9$$

x = _____

True or false?

The solution of $4w = 30$ is $w = 26$.

A. True

B. False

Find an unknown side length from the perimeter of a rectangle

MAIN

Grade 4 main question

4.MD.A.3

BACKWARD 1

Find an unknown side from the perimeter

3.MD.D.8

BACKWARD 2

Find an unknown addend within 100

2.NBT.B.5

FORWARD 1

Find an unknown side length from a perimeter with fractions

5.NF.A.2

FORWARD 2

Write and solve one-step equations for a missing side length

6.EE.B.7

SET 69 • 4.MD.A.3

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.A.3

A rectangle has a perimeter of 30 cm and a length of 10 cm.

What is its width?

Width: _____

True or false?

A square has a perimeter of 36 inches, so each side is 9 inches long.

A. True

B. False

A rectangular garden has a perimeter of 50 m. Its length is 15 m.

What is its width?

A. 20 m

B. 10 m

C. 25 m

D. 35 m

A rectangle has a perimeter of 64 inches and a width of 12 inches.

Write an equation with a letter for the length. Then solve it.

Equation: _____

Length: _____

True or false?

A rectangle has a perimeter of 26 feet and a length of 8 feet, so its width is 18 feet.

A. True

B. False

SET 69 • 4.MD.A.3

BACKWARD 1

Find an unknown side from the perimeter • Grade 3 • Measurement & Data •
3.MD.D.8

A triangle has a perimeter of 20 cm. Its two known sides add to 14 cm.

How long is the third side?

Length: _____

True or false?

A square with a perimeter of 12 m has sides that are 3 m long.

- A. True
- B. False

A rectangle has a perimeter of 14 ft. Its two lengths and one width add to 12 ft.

How long is the other width?

A. 2 ft

B. 7 ft

C. 9 ft

D. 4 ft

A rectangle has a perimeter of 18 in. Its two widths and one length add to 13 in.

How long is the other length?

Length: _____

True or false?

A square with a perimeter of 20 units has sides that are 10 units long.

- A. True
- B. False

SET 69 • 4.MD.A.3

BACKWARD 2

Find an unknown addend within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Find the unknown number.

$$20 + ? = 30$$

Unknown: _____

True or false?

If $24 + n = 64$, then $n = 40$.

A. True

B. False

$$? + 30 = 50$$

A. 30

B. 25

C. 80

D. 20

Find the unknown number.

$$? + 16 = 26$$

Unknown: _____

True or false?

If $36 + n = 50$, then $n = 24$.

A. True

B. False

SET 69 • 4.MD.A.3

FORWARD 1

Find an unknown side length from a perimeter with fractions • Grade 5 •
Number & Operations–Fractions • 5.NF.A.2

A rectangle has a perimeter of 5 m. Its length is $1\frac{1}{2}$ m.

What is its width?

Width: _____

True or false?

A square has a perimeter of 3 feet, so each side is $\frac{3}{4}$ foot long.

- A.** True
- B.** False

A rectangle has a perimeter of 7 inches. Its width is $1\frac{1}{4}$ inches.

What is its length?

A. $1\frac{1}{4}$ inches

B. $5\frac{3}{4}$ inches

C. $2\frac{1}{4}$ inches

D. $4\frac{1}{2}$ inches

A triangle has a perimeter of 2 yd. Two of its sides are $\frac{1}{2}$ yd and $\frac{2}{3}$ yd long.

How long is the third side?

Length: _____

True or false?

A rectangle has a perimeter of 4 m and a length of $\frac{3}{4}$ m, so its width is $1\frac{1}{2}$ m.

A. True

B. False

SET 69 • 4.MD.A.3

FORWARD 2

Write and solve one-step equations for a missing side length • Grade 6 •
Expressions & Equations • 6.EE.B.7

A square has a perimeter of 18.4 cm.

Write and solve an equation for its side length s .

Equation: _____

$s =$ _____

True or false?

A rectangle has a perimeter of 30 m, and its two widths total 12 m. Solving $2w = 12$ gives $w = 6$ m.

A. True

B. False

A triangle has a perimeter of 15.5 in., and two of its sides total 9.25 in.

Solve $x + 9.25 = 15.5$ for the third side x .

A. 24.75 in.

B. 6.75 in.

C. 6.25 in.

D. 5.25 in.

Solve for w.

$$2w = 13$$

$$w = \underline{\hspace{10cm}}$$

True or false?

A regular hexagon has a perimeter of 27 feet.
Solving $6s = 27$ gives $s = 21$ feet.

A. True

B. False

Make a line plot of measurements in fractions of a unit ($\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{8}$)

MAIN	Grade 4 main question	4.MD.B.4
BACKWARD 1	Read a line plot with halves and fourths	3.MD.B.4
BACKWARD 2	Line plots of whole-number lengths	2.MD.D.9
FORWARD 1	Make line plots of data given in halves, fourths, and eighths toge...	5.MD.B.2
FORWARD 2	Display numerical data in dot plots	6.SP.B.4

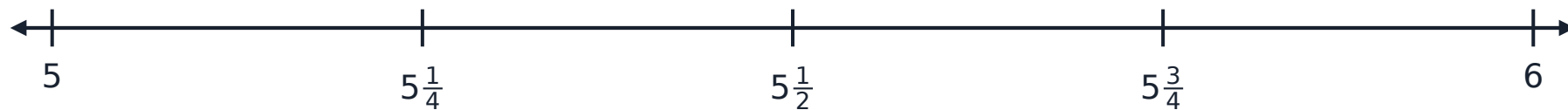
SET 50 • 4.MD.B.4

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.B.4

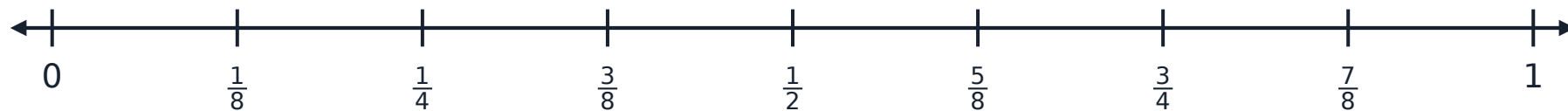
Pencil lengths (inches): $5\frac{1}{4}$, $5\frac{1}{2}$, $5\frac{1}{4}$, 6, $5\frac{3}{4}$, $5\frac{1}{2}$, $5\frac{1}{4}$

Make a line plot of the data.



Seedling heights (inches): $\frac{1}{8}$, $\frac{3}{8}$, $\frac{3}{8}$, $\frac{1}{2}$, $\frac{5}{8}$, $\frac{3}{8}$, $\frac{1}{8}$

Make a line plot of the data.



Shell lengths (inches): $\frac{1}{2}$, $\frac{3}{4}$, $\frac{1}{2}$, 1, $\frac{3}{4}$, $\frac{1}{2}$

In a line plot of the data, how many X's go above $\frac{1}{2}$?

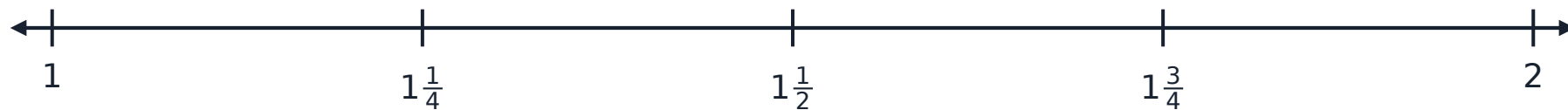
X's: _____

Which number line is best for a line plot of $\frac{1}{8}$, $\frac{3}{8}$, $\frac{5}{8}$, and $\frac{7}{8}$ inch?

- A.** A number line from 1 to 2 marked in fourths
- B.** A number line from 0 to 1 marked in halves
- C.** A number line from 0 to 1 marked in eighths
- D.** A number line from 0 to 8 marked in ones

Ribbon pieces (yards): $1\frac{1}{2}$, 2, $1\frac{1}{4}$, $1\frac{1}{2}$, $1\frac{3}{4}$, $1\frac{1}{2}$

Make a line plot of the data.



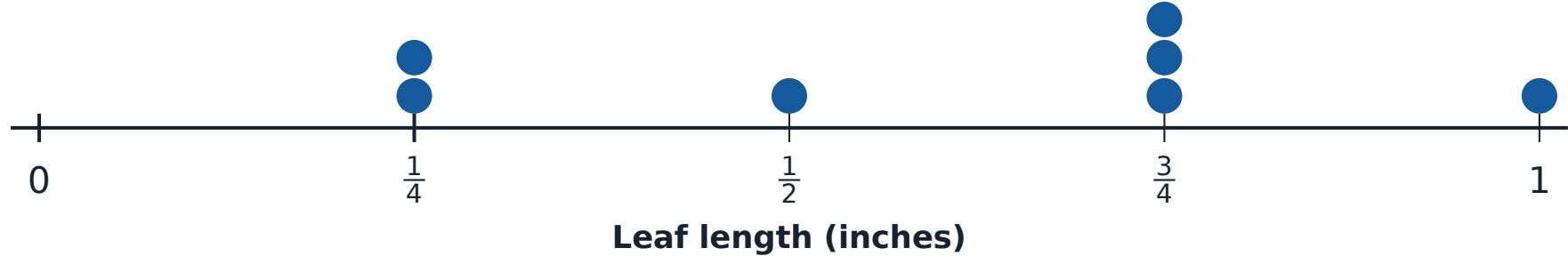
SET 50 • 4.MD.B.4

BACKWARD 1

Read a line plot with halves and fourths • Grade 3 • Measurement & Data •
3.MD.B.4

The line plot shows leaf lengths.

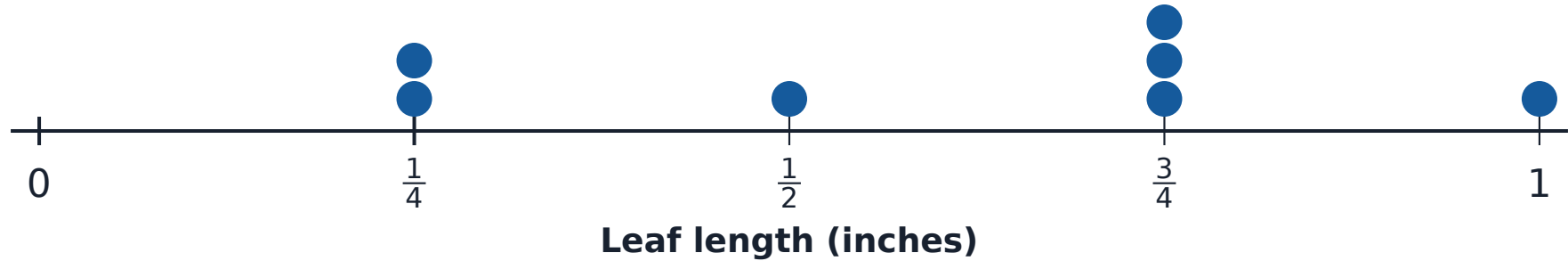
How many leaves are $\frac{3}{4}$ inch long?



Leaves: _____

True or false?

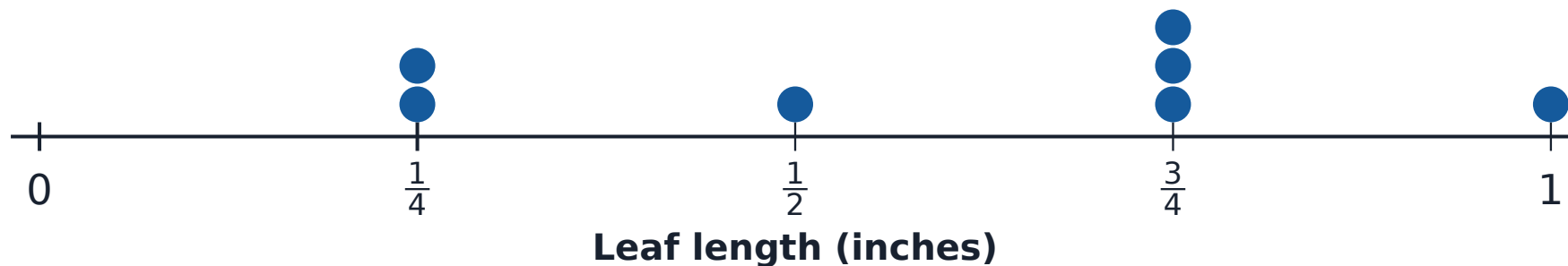
Two leaves are $\frac{1}{4}$ inch long.



A. True

B. False

Which leaf length is most common?



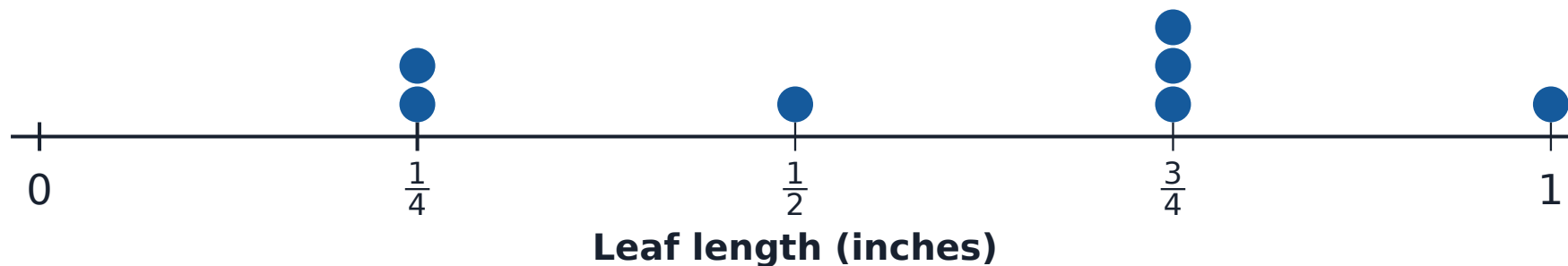
A. $\frac{1}{4}$ inch

B. $\frac{1}{2}$ inch

C. $\frac{3}{4}$ inch

D. 1 inch

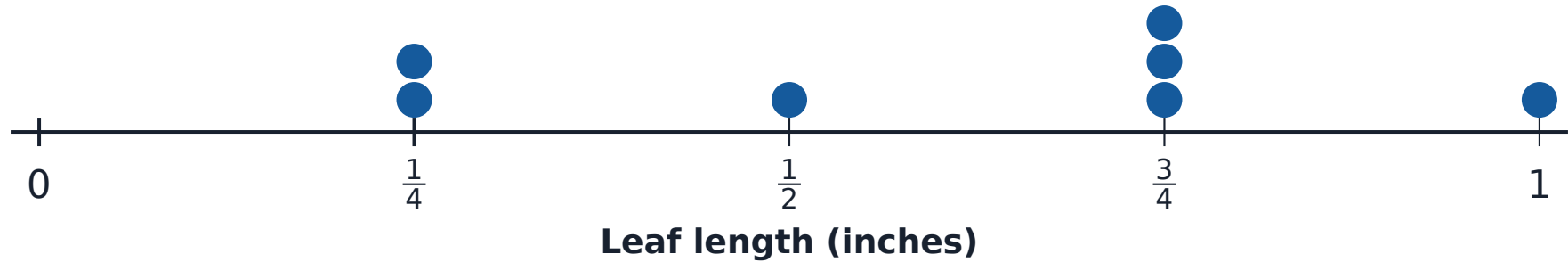
How many leaves were measured?



Leaves: _____

True or false?

No leaf is 1 inch long.



A. True

B. False

SET 50 • 4.MD.B.4

BACKWARD 2

Line plots of whole-number lengths • Grade 2 • Measurement & Data •
2.MD.D.9

Lengths (cm): 4, 6, 6, 7, 6

In a line plot of the data, how many X's go above 6?

X's: _____

True or false?

A line plot shows each measurement as a mark above a number line.

A. True

B. False

Which number line fits a line plot of 2, 3, 3, and 8?

- A.** A number line from 10 to 20
- B.** A number line from 0 to 3
- C.** A number line from 0 to 10
- D.** A number line from 4 to 6

Lengths (inches): 3, 5, 5, 6

Make a line plot of the data.



True or false?

A line plot of 1, 2, 2, 2 has 2 X's above 2.

- A. True
- B. False

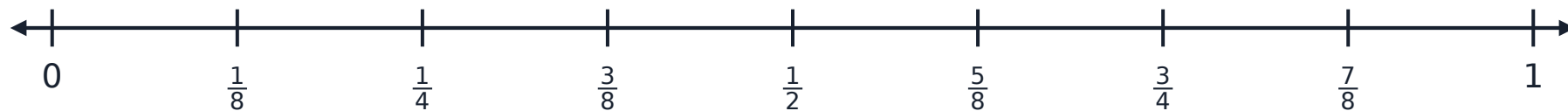
SET 50 • 4.MD.B.4

FORWARD 1

Make line plots of data given in halves, fourths, and eighths together • Grade 5
• Measurement & Data • 5.MD.B.2

Bead lengths (inches): $\frac{1}{2}$, $\frac{3}{8}$, $\frac{3}{4}$, $\frac{3}{8}$, $\frac{1}{2}$, $\frac{7}{8}$

Make a line plot of the data.



True or false?

A number line marked in eighths can show $\frac{1}{2}$, $\frac{1}{4}$, and $\frac{3}{8}$ inch on one line plot.

A. True

B. False

Data (pounds): $\frac{1}{4}$, $\frac{5}{8}$, $\frac{1}{2}$, $\frac{1}{4}$

In a line plot marked in eighths, how many X's go above $\frac{2}{8}$?

A. 2

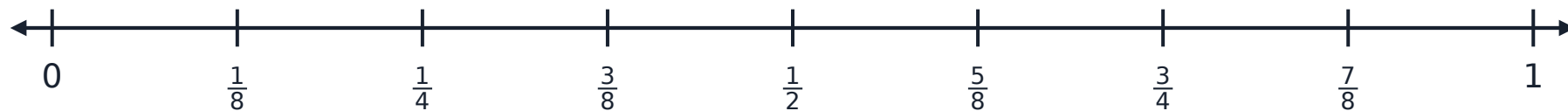
B. 1

C. 4

D. 0

Rain (inches): $\frac{1}{8}$, $\frac{1}{4}$, $\frac{1}{4}$, $\frac{3}{8}$, $\frac{1}{2}$

Make a line plot of the data.



True or false?

In a line plot of $\frac{1}{2}$, $\frac{2}{4}$, and $\frac{4}{8}$ cup, the three X's go above three different marks.

- A. True
- B. False

SET 50 • 4.MD.B.4

FORWARD 2

Display numerical data in dot plots • Grade 6 • Statistics & Probability •
6.SP.B.4

Number of pets: 0, 1, 1, 2, 2, 2, 3, 5

Make a dot plot of the data.



True or false?

A histogram groups data values into intervals.

A. True

B. False

Which display shows every individual data value?

- A.** A histogram
- B.** A table of intervals
- C.** A dot plot
- D.** A box plot

Data: 12, 15, 15, 18, 20

In a dot plot of the data, how many dots go above 15?

Dots: _____

True or false?

A dot plot cannot show data with repeated values.

A. True

B. False

Solve addition and subtraction problems using fraction data in a line plot

MAIN	Grade 4 main question	4.MD.B.4
BACKWARD 1	Read a line plot of lengths in fourths of an inch	3.MD.B.4
BACKWARD 2	Equivalent halves, fourths, and eighths	3.NF.A.3.b
FORWARD 1	Use operations on fractions to solve line plot problems	5.MD.B.2
FORWARD 2	Find the mean of a data set	6.SP.B.5.c

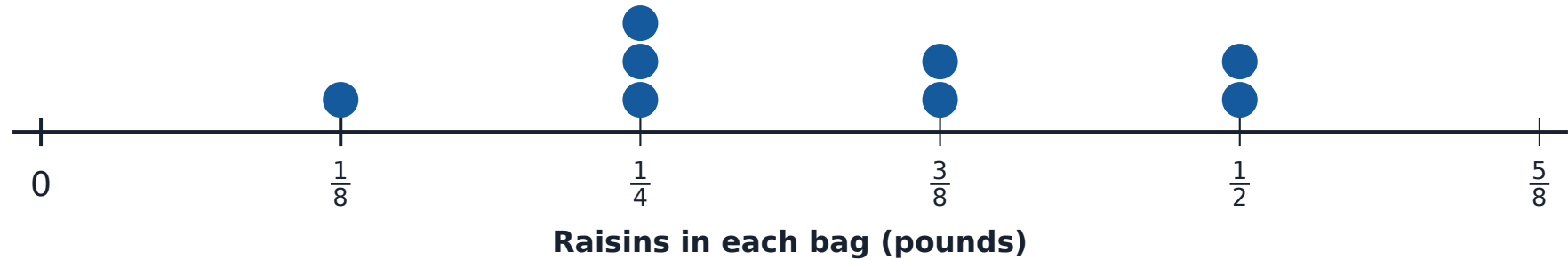
SET 51 • 4.MD.B.4

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.B.4

The line plot shows the raisins in some bags.

What is the difference between the heaviest and the lightest bag?



Difference: _____

What is the total weight of the bags that weigh $\frac{1}{4}$ pound each?



Total: _____

How many bags weigh more than $\frac{1}{4}$ pound?



A. 2

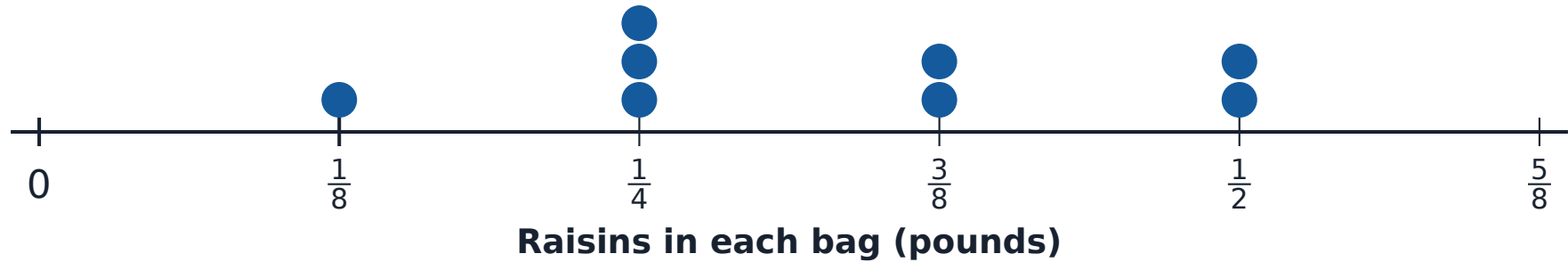
B. 3

C. 5

D. 4

True or false?

The two $\frac{1}{2}$ -pound bags weigh 1 pound in all.



A. True

B. False

What is the total weight of all the bags?



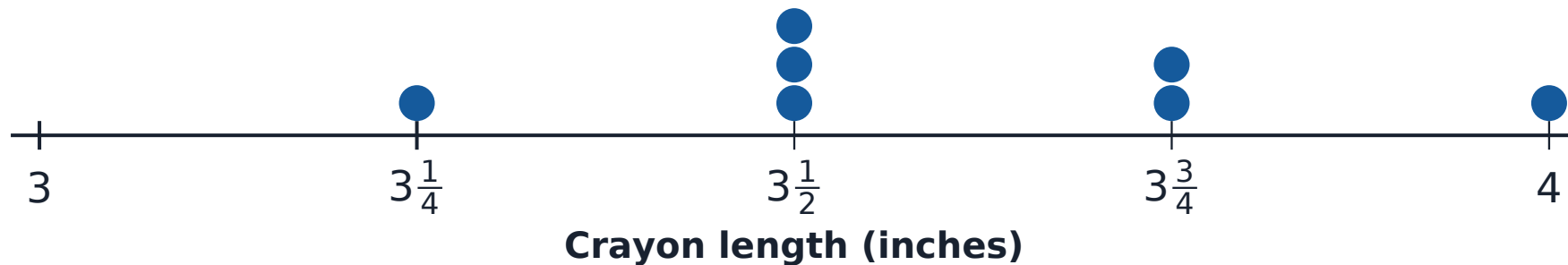
Total: _____

SET 51 • 4.MD.B.4

BACKWARD 1

Read a line plot of lengths in fourths of an inch • Grade 3 • Measurement & Data • 3.MD.B.4

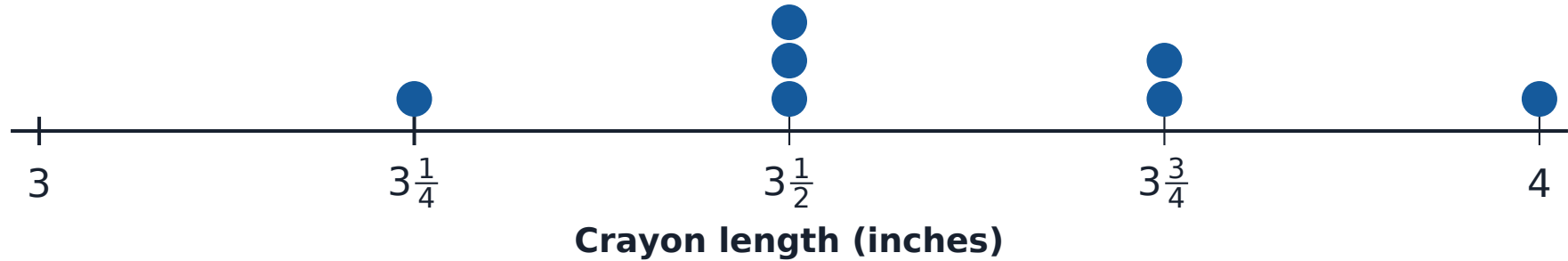
How many crayons are $3\frac{1}{2}$ inches long?



Crayons: _____

True or false?

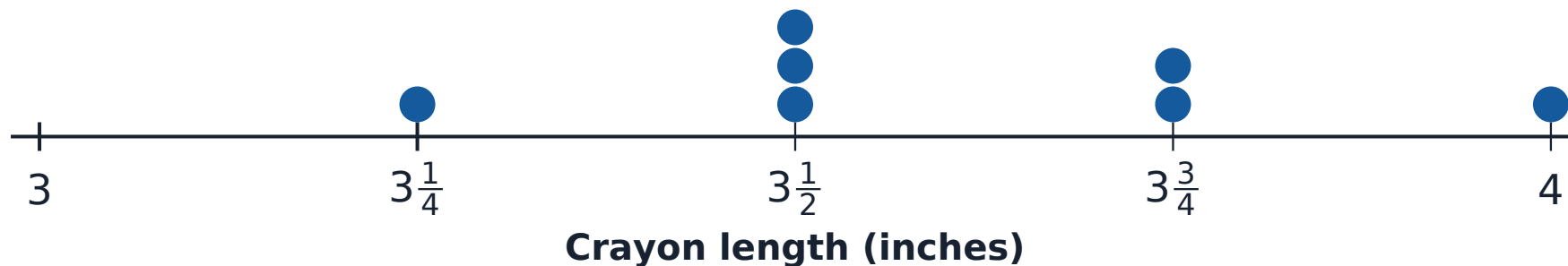
One crayon is 4 inches long.



A. True

B. False

Which length do exactly 2 crayons have?



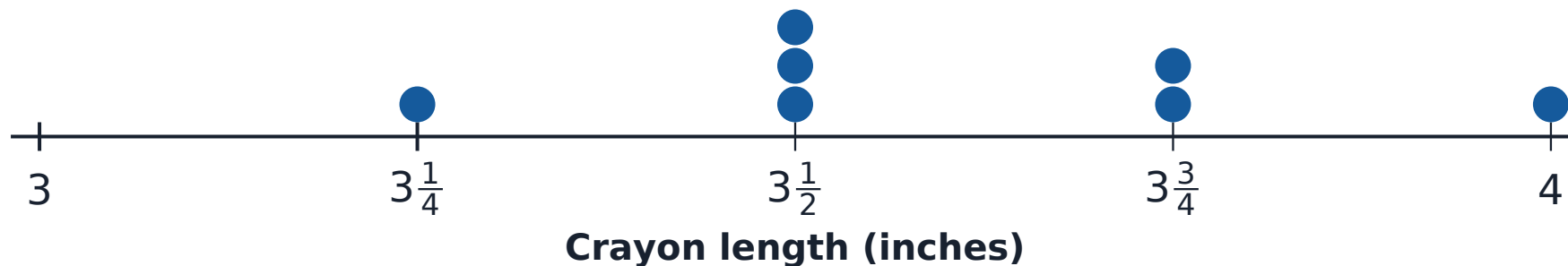
A. $3\frac{1}{2}$
inches

B. 4 inches

C. $3\frac{1}{4}$
inches

D. $3\frac{3}{4}$
inches

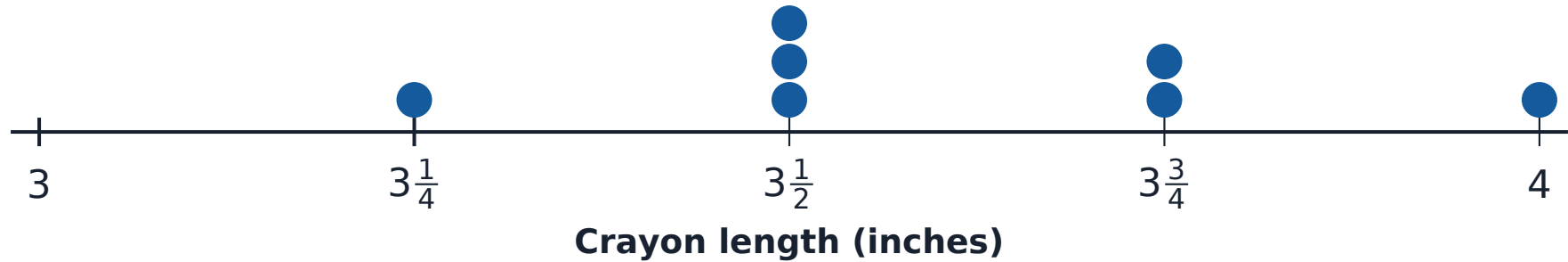
How many crayons were measured?



Crayons: _____

True or false?

The shortest crayon is 3 inches long.



A. True

B. False

SET 51 • 4.MD.B.4

BACKWARD 2

Equivalent halves, fourths, and eighths • Grade 3 • Number & Operations–Fractions • 3.NF.A.3.b

Write $\frac{1}{2}$ as eighths.

Fraction: _____

True or false?

$$\frac{1}{4} = \frac{2}{8}$$

A. True

B. False

Which fraction is equal to $\frac{3}{4}$?

A. $\frac{5}{8}$

B. $\frac{6}{8}$

C. $\frac{4}{8}$

D. $\frac{3}{8}$

Find the missing number.

$$\frac{1}{2} = \frac{?}{4}$$

Missing number: _____

True or false?

$$\frac{2}{8} = \frac{1}{2}$$

A. True

B. False

SET 51 • 4.MD.B.4

FORWARD 1

Use operations on fractions to solve line plot problems • Grade 5 •
Measurement & Data • 5.MD.B.2

Juice in 3 glasses (cups): $\frac{1}{2}$, $\frac{3}{4}$, $1\frac{1}{4}$

If the juice were shared equally, how much would each glass hold?

Cups: _____

True or false?

Water in beakers (liters): $\frac{1}{8}$, $\frac{1}{4}$, $\frac{3}{8}$. The total is $\frac{3}{4}$ liter.

- A. True
- B. False

Ribbons (yards): $\frac{1}{2}$, $\frac{1}{2}$, $\frac{3}{4}$, $\frac{1}{4}$

What is the total length?

A. $1\frac{3}{4}$ yards

B. $2\frac{1}{4}$ yards

C. $1\frac{1}{2}$ yards

D. 2 yards

Rock masses (kg): $\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{2}$, $\frac{3}{4}$

If the total mass were shared equally by the 4 rocks, what would each rock's mass be?

Mass: _____

True or false?

The total of $\frac{1}{2}$, $\frac{1}{4}$, and $\frac{1}{8}$ is $\frac{3}{14}$.

A. True

B. False

SET 51 • 4.MD.B.4

FORWARD 2

Find the mean of a data set • Grade 6 • Statistics & Probability • 6.SP.B.5.c

Find the mean.

4, 6, 8, 10

Mean: _____

True or false?

The mean of 2, 4, and 9 is 5.

A. True

B. False

Find the mean.

$$\frac{1}{2}, \frac{1}{2}, 1, 2$$

A. 4

B. $\frac{1}{2}$

C. 2

D. 1

Five plants grew 3, 5, 5, 6, and 6 cm.

What is the mean growth?

Mean: _____

True or false?

The mean of 10, 10, and 40 is 10.

A. True

B. False

An angle is measured as a fraction of a full circle; a one-degree angle is $\frac{1}{360}$ of a circle

MAIN	Grade 4 main question	4.MD.C.5.a
BACKWARD 1	Equal parts of a shape	3.G.A.2
BACKWARD 2	Turns of the minute hand on a clock	3.MD.A.1
FORWARD 1	Classify triangles by their angles	5.G.B.4
FORWARD 2	Find a percent of a full turn — nearest related (no direct Grade 6 s...	6.RP.A.3.c

SET 52 • 4.MD.C.5.a

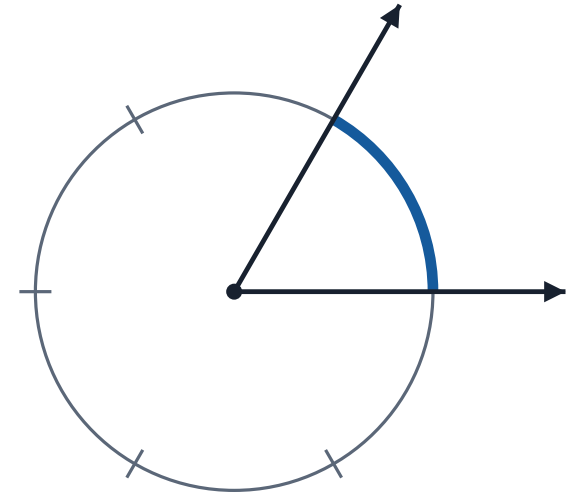
MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.C.5.a

The circle is centered where the rays meet. The thick arc is $\frac{1}{6}$ of the circle.

What is the angle's measure? Explain.

What angle has an arc of $\frac{1}{360}$?



Measure: _____

Explanation: _____

Arc of $\frac{1}{360}$: _____

True or false?

An angle that turns through $\frac{1}{4}$ of a full circle measures 90° .

- A.** True
- B.** False

What fraction of a full circle does a 180° angle turn through?

A. $\frac{1}{3}$

B. $\frac{1}{2}$

C. $\frac{1}{180}$

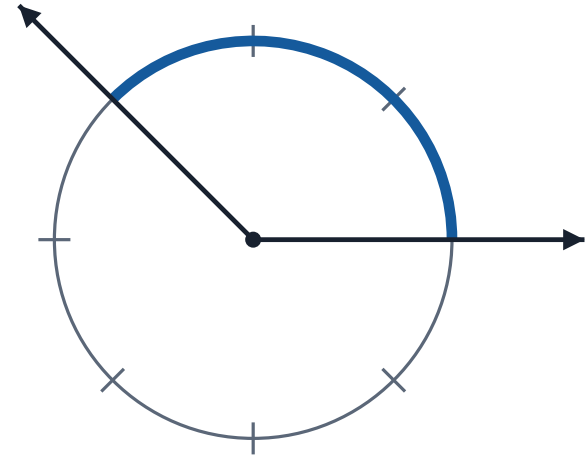
D. $\frac{1}{4}$

The circle is centered at the common endpoint of the two rays and is divided into 8 equal arcs. and is divided into 8 equal arcs.

What fraction of the circle is the thick arc between the rays? What is the measure of the angle?

Fraction: _____

Measure: _____



True or false?

An angle that turns through $\frac{1}{3}$ of a full circle measures 30° .

- A.** True
- B.** False

SET 52 • 4.MD.C.5.a

BACKWARD 1

Equal parts of a shape • Grade 3 • Geometry • 3.G.A.2

A circle is cut into 4 equal parts.

What fraction of the circle is 1 part?

Fraction: _____

True or false?

A circle cut into 2 equal parts is cut into halves.

A. True

B. False

A circle is cut into 6 equal parts.

What fraction of the circle is each part?

A. $\frac{1}{3}$

B. $\frac{5}{6}$

C. $\frac{6}{1}$

D. $\frac{1}{6}$

Each part of a circle is $\frac{1}{8}$ of the circle.

Into how many equal parts is the circle cut?

Parts: _____

True or false?

A circle cut into 3 parts of different sizes is cut into thirds.

A. True

B. False

SET 52 • 4.MD.C.5.a

BACKWARD 2

Turns of the minute hand on a clock • Grade 3 • Measurement & Data •
3.MD.A.1

The minute hand turns from 12 to 3.
How many minutes pass?

Minutes: _____

True or false?

When the minute hand makes a full turn, 60 minutes pass.

- A.** True
- B.** False

The minute hand turns halfway around the clock.

How many minutes pass?

- A.** 60
- B.** 30
- C.** 15
- D.** 12

The minute hand turns from 12 to 9.
How many minutes pass?

Minutes: _____

True or false?

A quarter turn of the minute hand takes 25 minutes.

A. True

B. False

SET 52 • 4.MD.C.5.a

FORWARD 1

Classify triangles by their angles • Grade 5 • Geometry • 5.G.B.4

A triangle has angles of 90° , 60° , and 30° .
Classify it by its angles.

Classification: _____

True or false?

A triangle with angles of 70° , 60° , and 50° is an acute triangle.

- A. True
- B. False

A triangle has one angle of 120° .

What kind of triangle is it?

- A.** Right
- B.** Obtuse
- C.** Acute
- D.** Equilateral

Can a triangle have two right angles? Write yes or no.

Answer: _____

True or false?

An obtuse triangle has three angles that are each less than 90° .

- A. True
- B. False

FORWARD 2

Find a percent of a full turn • Grade 6 • Ratios & Proportional Relationships •
6.RP.A.3.c

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

What is 25% of 360° ?

Answer: _____

True or false?

50% of a full turn is 180° .

A. True

B. False

What percent of a full circle is 36° ?

- A.** 10%
- B.** 1%
- C.** 20%
- D.** 36%

What is 75% of 360° ?

Answer: _____

True or false?

20% of 360° is 20° .

A. True

B. False

An angle that turns through n one-degree angles has a measure of n degrees

MAIN	Grade 4 main question	4.MD.C.5.b
BACKWARD 1	A shape covered by n unit squares has an area of n square units	3.MD.C.5.b
BACKWARD 2	Measure length by laying units end to end	1.MD.A.2
FORWARD 1	A solid packed with n unit cubes has a volume of n cubic units —...	5.MD.C.3.b
FORWARD 2	Find volume by packing with unit-fraction cubes — nearest relate...	6.G.A.2

SET 53 • 4.MD.C.5.b

MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.C.5.b

An angle turns through 45 one-degree angles.
What is its measure?

Measure: _____

True or false?

An angle made of 120 one-degree angles measures 120° .

A. True

B. False

How many one-degree angles make a 75° angle?

- A.** 15
- B.** 360
- C.** 75
- D.** 285

A fan blade turns through 30 one-degree angles and then 20 more in the same direction.

What angle did it turn through?

Angle: _____

True or false?

A 90° angle is made of 9 one-degree angles.

A. True

B. False

SET 53 • 4.MD.C.5.b

BACKWARD 1

A shape covered by n unit squares has an area of n square units • Grade 3 •
Measurement & Data • 3.MD.C.5.b

A shape is covered by 14 unit squares with no gaps or overlaps.

What is its area?

Area: _____

True or false?

A shape covered by 9 unit squares with no gaps has an area of 9 square units.

- A.** True
- B.** False

A rectangle is covered by 20 unit squares.

What is its area?

- A.** 10 square units
- B.** 20 units
- C.** 40 square units
- D.** 20 square units

How many unit squares cover a shape with an area of 6 square units?

Unit squares: _____

True or false?

A shape covered by 8 unit squares has an area of 16 square units.

- A.** True
- B.** False

SET 53 • 4.MD.C.5.b

BACKWARD 2

Measure length by laying units end to end • Grade 1 • Measurement & Data •
1.MD.A.2

A pencil is as long as 6 paper clips laid end to end.
How many paper clips long is it?

Paper clips: _____

True or false?

Length units laid end to end must have no gaps or overlaps.

A. True

B. False

A crayon is as long as 4 cubes in a row.

How long is the crayon?

- A.** 5 cubes
- B.** 4 cubes
- C.** 1 cube
- D.** 8 cubes

A book is 9 tiles long. A desk is 3 tiles longer.
How many tiles long is the desk?

Tiles: _____

True or false?

If the cubes overlap when you measure, the count is still correct.

- A.** True
- B.** False

FORWARD 1

A solid packed with n unit cubes has a volume of n cubic units • Grade 5 •
Measurement & Data • 5.MD.C.3.b

► No Grade 5 standard directly continues this skill. This branch uses the nearest related Grade 5 standard.

A box is filled with 24 unit cubes with no gaps or overlaps.

What is its volume?

Volume: _____

True or false?

A solid built from 15 unit cubes has a volume of 15 cubic units.

- A.** True
- B.** False

A prism has 3 layers with 8 unit cubes in each layer.

What is its volume?

- A.** 24 square units
- B.** 11 cubic units
- C.** 24 cubic units
- D.** 38 cubic units

How many unit cubes fill a box with a volume of 40 cubic units?

Unit cubes: _____

True or false?

A cube with edges 1 unit long has a volume of 6 cubic units.

A. True

B. False

FORWARD 2

Find volume by packing with unit-fraction cubes • Grade 6 • Geometry • 6.G.A.2

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

How many cubes with edges $\frac{1}{2}$ unit long fill a cube with edges 1 unit long?

Cubes: _____

True or false?

A cube with edges $\frac{1}{2}$ unit long has a volume of $\frac{1}{8}$ cubic unit.

- A. True
- B. False

A box is packed with 16 cubes that each have edges $\frac{1}{2}$ unit long.

What is its volume?

A. 2 cubic units

B. 16 cubic units

C. 4 cubic units

D. 8 cubic units

Find the volume of a prism that is 2 units by $1\frac{1}{2}$ units by $\frac{1}{2}$ unit.

Volume: _____

True or false?

A box that is 1 unit by 1 unit by $\frac{1}{2}$ unit holds exactly 2 cubes with edges $\frac{1}{2}$ unit long.

A. True

B. False

Measure angles in whole-number degrees using a protractor

MAIN	Grade 4 main question	4.MD.C.6
BACKWARD 1	Measure length with a ruler	2.MD.A.1
BACKWARD 2	Shapes with a given number of angles	2.G.A.1
FORWARD 1	Attributes of a category belong to its subcategories	5.G.B.3
FORWARD 2	Write and solve $x + p = q$ to find an unknown angle — nearest re...	6.EE.B.7

SET 54 • 4.MD.C.6

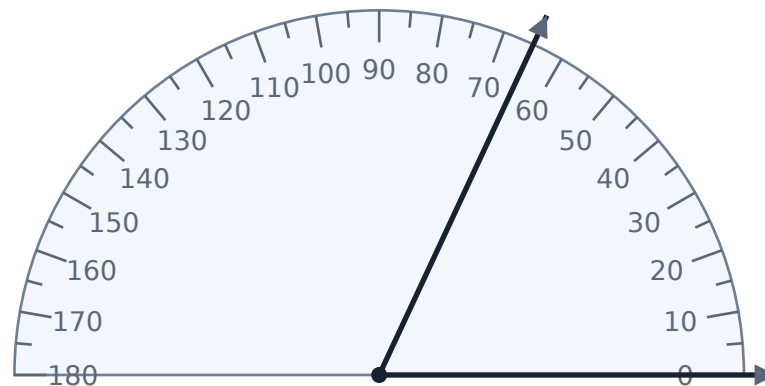
MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.C.6

Use the protractor.

What is the measure of the angle?

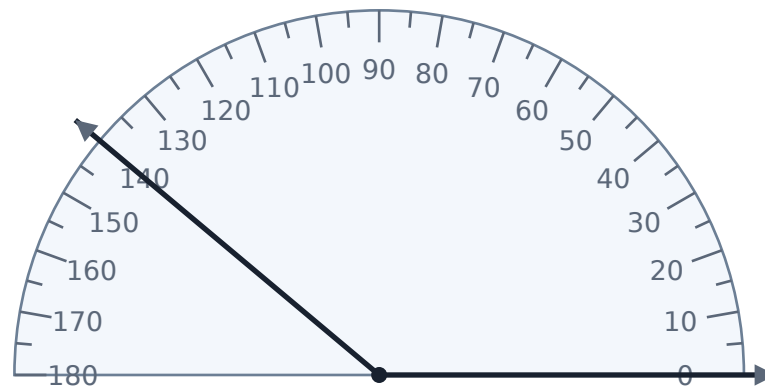
Measure: _____



Use the protractor.

What is the measure of the angle?

Measure: _____



Use the protractor. One ray points to 180 on the outer scale.

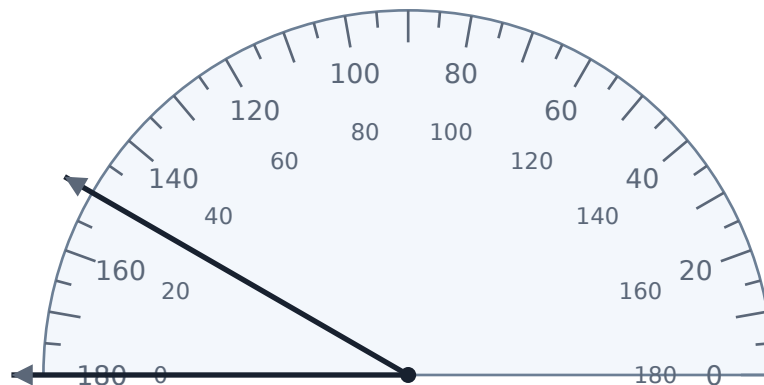
What is the measure of the angle?

A. 150°

B. 120°

C. 180°

D. 30°

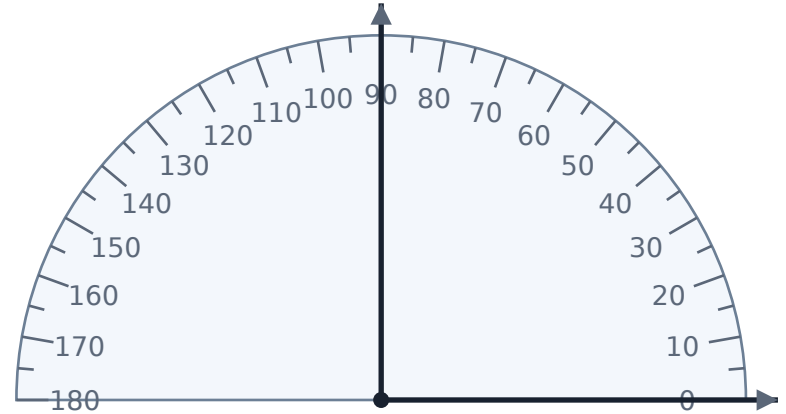


True or false?

The angle measures 90° .

A. True

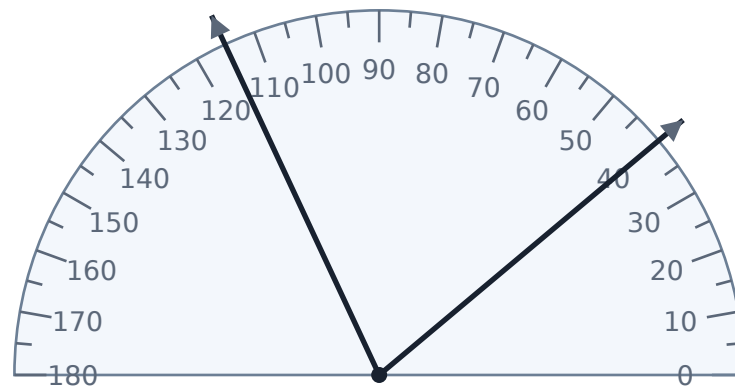
B. False



Use the protractor.

What is the measure of the angle?

Measure: _____



SET 54 • 4.MD.C.6

BACKWARD 1

Measure length with a ruler • Grade 2 • Measurement & Data • 2.MD.A.1

A pencil reaches from 0 to 7 on an inch ruler.

How long is the pencil?

Length: _____

True or false?

When you measure with a ruler, you can line up one end of the object with 0.

- A.** True
- B.** False

A crayon starts at 2 cm and ends at 9 cm on a ruler.

How long is the crayon?

- A.** 2 cm
- B.** 7 cm
- C.** 9 cm
- D.** 11 cm

A leaf reaches from 0 to 5 on a centimeter ruler.
How long is the leaf?

Length: _____

True or false?

An eraser that reaches from 3 to 8 on a ruler is 8 units long.

- A.** True
- B.** False

SET 54 • 4.MD.C.6

BACKWARD 2

Shapes with a given number of angles • Grade 2 • Geometry • 2.G.A.1

How many angles does a triangle have?

Angles: _____

True or false?

A square has 4 angles.

A. True

B. False

Which shape has 6 angles?

- A.** Pentagon
- B.** Quadrilateral
- C.** Hexagon
- D.** Triangle

How many angles does a pentagon have?

Angles: _____

True or false?

A circle has 4 angles.

A. True

B. False

SET 54 • 4.MD.C.6

FORWARD 1

Attributes of a category belong to its subcategories • Grade 5 • Geometry •
5.G.B.3

True or false?

Every right triangle has one 90° angle. An isosceles right triangle is a right triangle, so it has a 90° angle.

A. True

B. False

A shape is a rhombus.

Which statement must be true?

- A.** Its opposite angles are equal.
- B.** All its angles are 90° .
- C.** It has 3 sides.
- D.** It has exactly one pair of parallel sides.

A quadrilateral has four 90° angles and four sides of equal length.

What is its most specific name?

Name: _____

True or false?

Every parallelogram has four 90° angles.

A. True

B. False

A triangle has angles of 45° , 45° , and 90° .

Name two categories of triangles it belongs to.

Categories: _____

FORWARD 2

Write and solve $x + p = q$ to find an unknown angle • Grade 6 • Expressions & Equations • 6.EE.B.7

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

Two angles together make a 90° angle. One of them is 35° .

Write and solve an equation for the other angle, x .

Equation: _____

$x =$ _____

True or false?

If $x + 120 = 180$, then $x = 60$.

A. True

B. False

Solve for a.

$$a + 48 = 90$$

A. 42

B. 138

C. 48

D. 52

Solve for y .

$$y + 75 = 180$$

$$y = \underline{\hspace{10cm}}$$

True or false?

If $2x = 90$, then $x = 180$.

A. True

B. False

Sketch angles of a given measure using a protractor

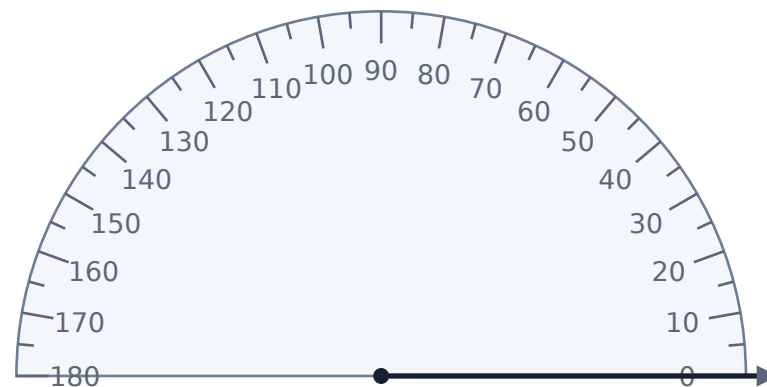
MAIN	Grade 4 main question	4.MD.C.6
BACKWARD 1	Draw shapes with given attributes	2.G.A.1
BACKWARD 2	Defining attributes of shapes	1.G.A.1
FORWARD 1	Graph points in the first quadrant to solve problems	5.G.A.2
FORWARD 2	Draw polygons in the coordinate plane	6.G.A.3

SET 55 • 4.MD.C.6

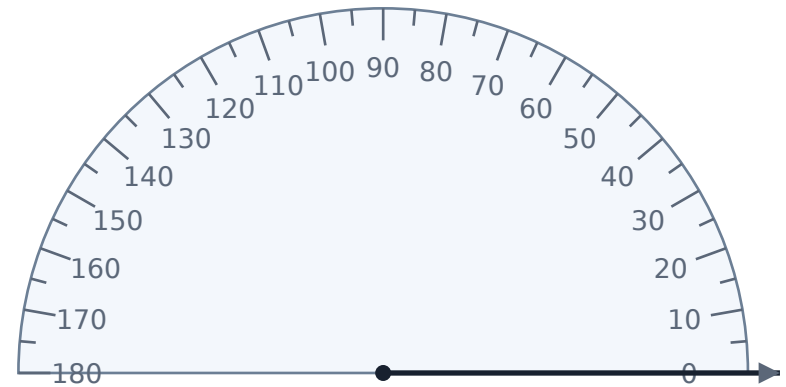
MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.C.6

Use the protractor to draw an angle of 50° . One ray is drawn for you.

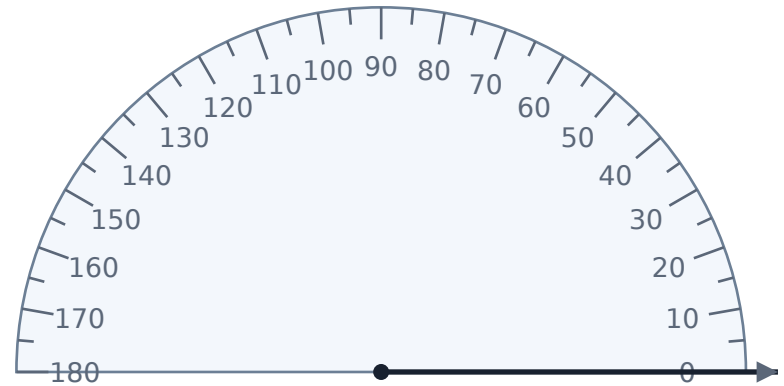


Use the protractor to draw an angle of 120° . One ray is drawn for you.



Use the protractor to draw an angle of 90° . One ray is drawn for you.

What kind of angle is it?

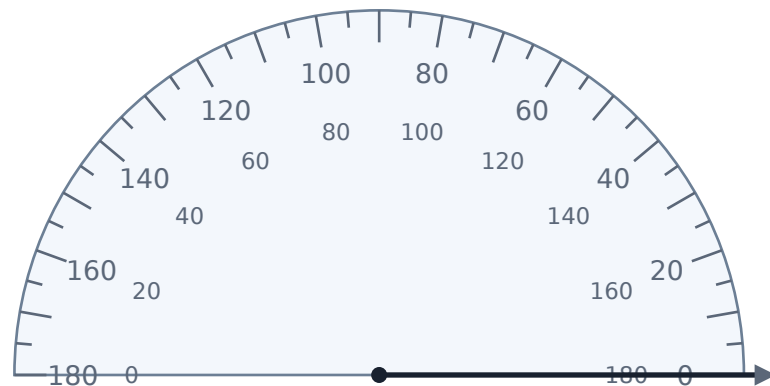


Kind of angle: _____

Kim wants to draw a 70° angle. Her first ray points to 0 on the outer scale.

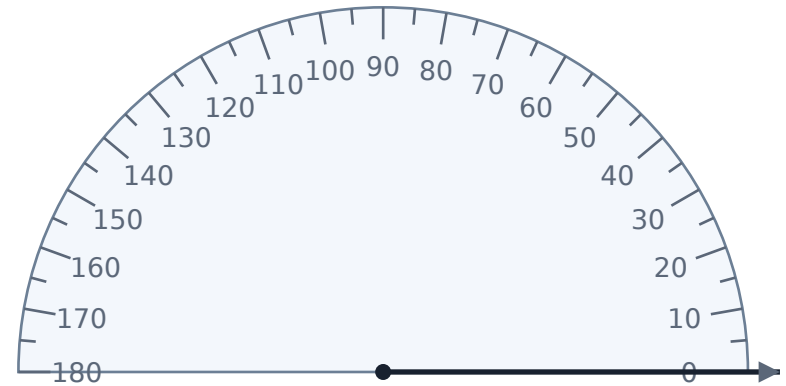
Where should she draw the second ray?

- A.** Through 70 on the inner scale
- B.** Through 0 on the inner scale
- C.** Through 110 on the outer scale
- D.** Through 70 on the outer scale



Use the protractor to draw an angle of 35° . One ray is drawn for you.

Is the angle acute or obtuse?



Answer: _____

SET 55 • 4.MD.C.6

BACKWARD 1

Draw shapes with given attributes • Grade 2 • Geometry • 2.G.A.1

A shape has 4 sides and 4 angles.

Name a shape it could be.

Shape: _____

True or false?

A triangle has 3 sides and 3 angles.

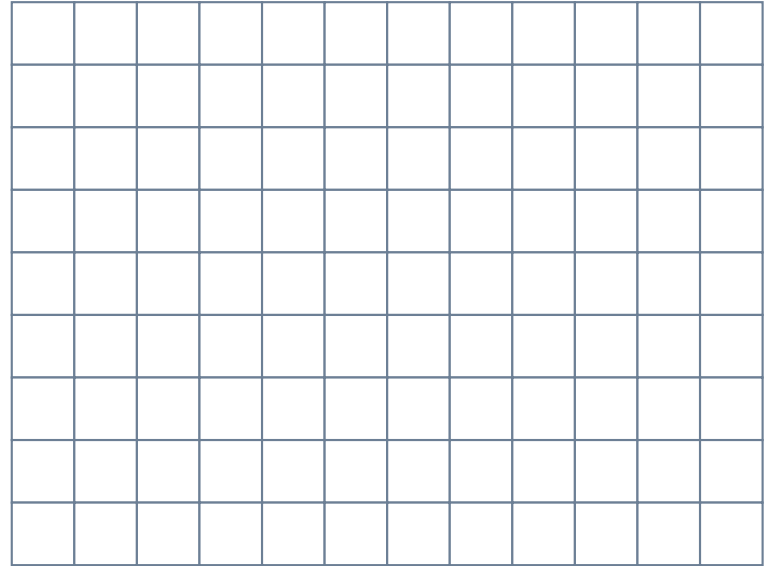
A. True

B. False

Which shape has 5 sides?

- A.** Triangle
- B.** Hexagon
- C.** Pentagon
- D.** Square

Draw a shape that has 3 sides.



True or false?

A shape with 6 angles has 4 sides.

A. True

B. False

SET 55 • 4.MD.C.6

BACKWARD 2

Defining attributes of shapes • Grade 1 • Geometry • 1.G.A.1

True or false?

A triangle can be any color or size and still be a triangle.

A. True

B. False

Which is a defining attribute of a square?

- A.** It is turned on its corner.
- B.** It is blue.
- C.** It is big.
- D.** It has 4 equal sides.

How many corners (vertices) does a rectangle have?

Corners: _____

True or false?

A shape must be large to be a hexagon.

A. True

B. False

A closed shape has 3 straight sides.

What is it called?

Name: _____

SET 55 • 4.MD.C.6

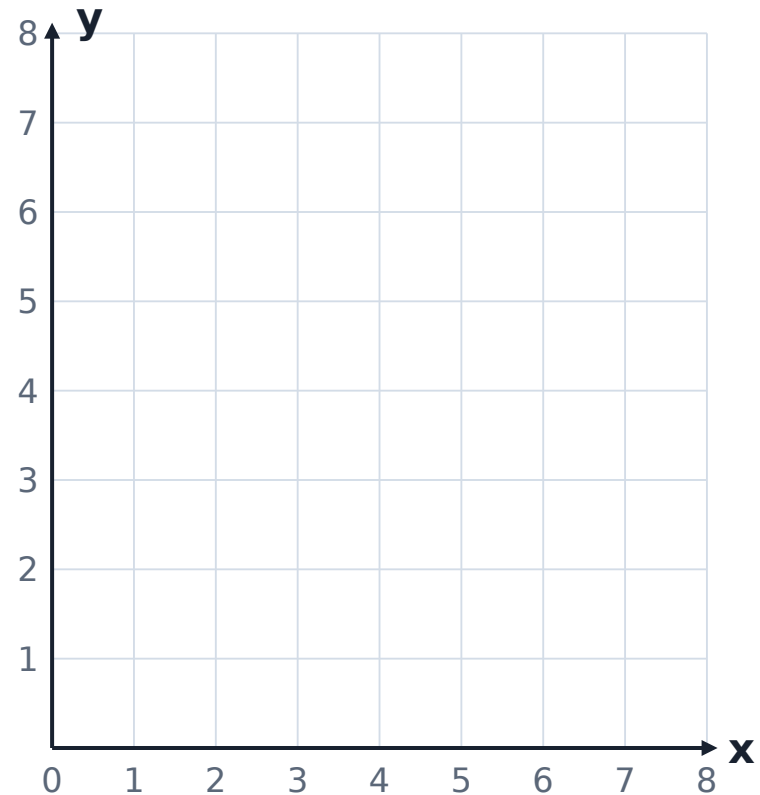
FORWARD 1

Graph points in the first quadrant to solve problems • Grade 5 • Geometry •
5.G.A.2

On a map grid, Ana's house is at (1, 2) and the park is at (1, 7). Each unit is 1 block.

Plot and label both points.
How many blocks apart are they?

Blocks: _____



True or false?

A plant is 4 cm tall on day 0 and grows 2 cm each day. The point (3, 10) shows its height on day 3.

A. True

B. False

A tank holds 2 liters at minute 0 and fills 3 liters each minute.

Which point (minutes, liters) shows the amount after 4 minutes?

A. (4, 14)

B. (3, 4)

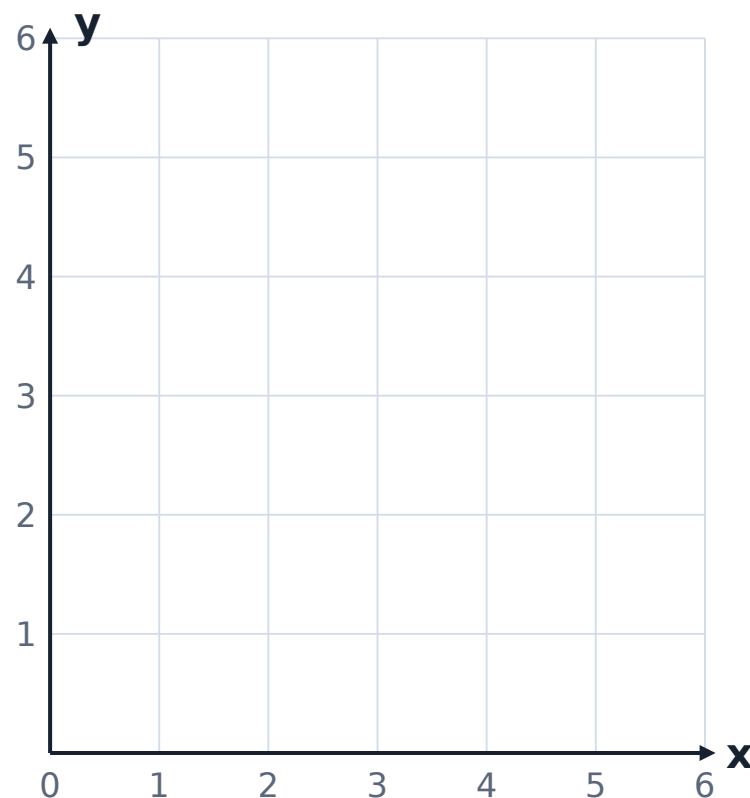
C. (4, 12)

D. (14, 4)

Plot $(1, 1)$, $(5, 1)$, and $(1, 4)$,
and connect them.

What kind of angle is at $(1, 1)$?

Kind of angle: _____



True or false?

On a map grid, a store is at (6, 3) and a bank is at (2, 3). They are 8 units apart.

A. True

B. False

SET 55 • 4.MD.C.6

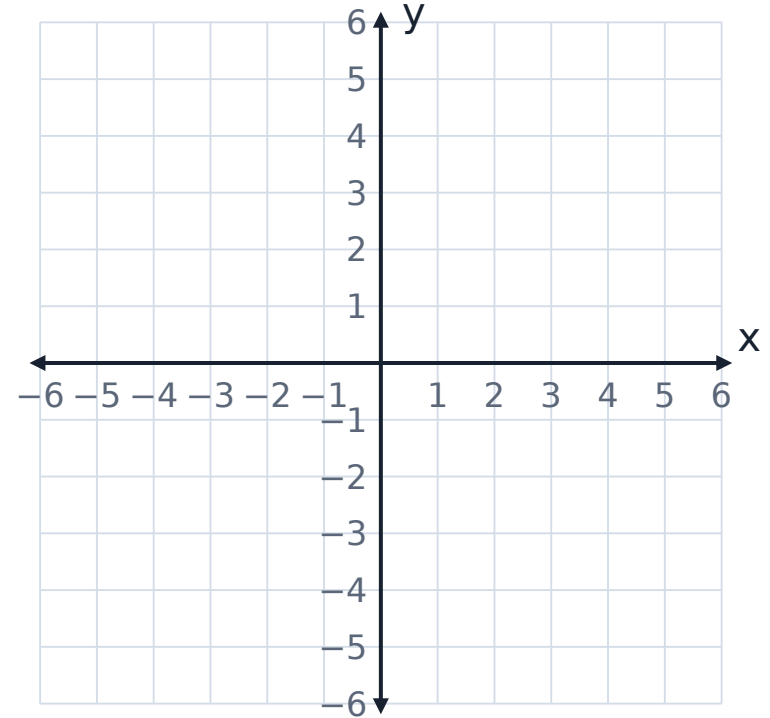
FORWARD 2

Draw polygons in the coordinate plane • Grade 6 • Geometry • 6.G.A.3

Plot $A(-3, 2)$, $B(3, 2)$, $C(3, -2)$, and $D(-3, -2)$. Connect them in order.

What is the most specific name for the polygon?

Name: _____



What is the length of the side from $(-2, 4)$ to $(5, 4)$?

Length: _____

Three vertices of a square are $(0, 0)$, $(4, 0)$, and $(4, 4)$.

What is the fourth vertex?

A. $(8, 4)$

B. $(0, 8)$

C. $(4, 8)$

D. $(0, 4)$

True or false?

The points $(1, 1)$, $(1, 5)$, and $(4, 1)$ are the vertices of a right triangle.

- A. True
- B. False

True or false?

The side from $(-3, -1)$ to $(2, -1)$ is 1 unit long.

A. True

B. False

Angle measure is additive

MAIN	Grade 4 main question	4.MD.C.7
BACKWARD 1	Add within 100	2.NBT.B.5
BACKWARD 2	Area is additive	3.MD.C.7.d
FORWARD 1	Volume is additive — nearest related (no direct Grade 5 step)	5.MD.C.5.c
FORWARD 2	Write expressions and equations for parts of an angle — nearest...	6.EE.B.6

SET 56 • 4.MD.C.7

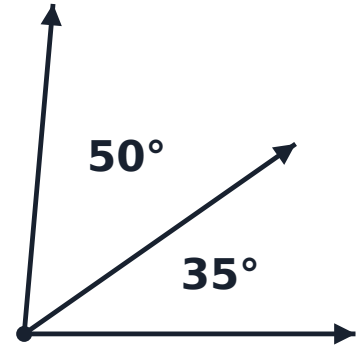
MAIN

Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.C.7

Angle ABC is split into two angles of 35° and 50° .

What is the measure of angle ABC?

Measure: _____



True or false?

A 90° angle is split into two angles. One is 40° , so the other is 50° .

A. True

B. False

Angles of 65° and 25° are side by side with no overlap.
What angle do they make together?

- A.** 90°
- B.** 100°
- C.** 80°
- D.** 40°

Angles of 30° , 45° , and 60° are side by side with no overlap.

What angle do they make together?

Measure: _____

True or false?

An angle of 120° can be split into angles of 70° and 60° .

A. True

B. False

SET 56 • 4.MD.C.7

BACKWARD 1

Add within 100 • Grade 2 • Number & Operations in Base Ten • 2.NBT.B.5

Add.

$$35 + 50$$

Sum: _____

True or false?

$$65 + 25 = 90$$

A. True

B. False

$$45 + 38 = ?$$

A. 713

B. 93

C. 73

D. 83

Add.

$$27 + 46$$

Sum: _____

True or false?

$$58 + 34 = 82$$

A. True

B. False

SET 56 • 4.MD.C.7

BACKWARD 2

Area is additive • Grade 3 • Measurement & Data • 3.MD.C.7.d

A figure is made of two rectangles that do not overlap.
Their areas are 12 and 15 square units.

What is the area of the figure?

Area: _____

True or false?

The area of a figure made of parts that do not overlap is the sum of the areas of the parts.

A. True

B. False

A figure is made of two rectangles that do not overlap.
Their areas are 12 and 6 square units.

What is the area of the figure?

- A.** 6 square units
- B.** 72 square units
- C.** 12 square units
- D.** 18 square units

Two squares that do not overlap have areas of 9 and 16 square units.

What is their total area?

Area: _____

True or false?

Two rectangles of 10 and 6 square units that do not overlap have a total area of 60 square units.

A. True

B. False

FORWARD 1

Volume is additive • Grade 5 • Measurement & Data • 5.MD.C.5.c

► No Grade 5 standard directly continues this skill. This branch uses the nearest related Grade 5 standard.

Two prisms that do not overlap have volumes of 24 and 36 cubic units.

What is their total volume?

Volume: _____

True or false?

A figure made of a 2-by-3-by-4 prism and a 1-by-3-by-4 prism has a volume of 36 cubic units.

- A. True
- B. False

A figure is made of a 5-by-2-by-2 prism and a 3-by-2-by-1 prism that do not overlap.

What is its volume?

- A.** 26 cubic units
- B.** 20 cubic units
- C.** 120 cubic units
- D.** 6 cubic units

A building is a 9-by-6-by-5 prism with a 3-by-6-by-2 prism attached.

What is its total volume?

Volume: _____

True or false?

The volume of two prisms that touch is the product of their volumes.

- A. True
- B. False

FORWARD 2

Write expressions and equations for parts of an angle • Grade 6 • Expressions & Equations • 6.EE.B.6

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

An angle is split into two parts, x° and 40° .

Write an expression for the measure of the whole angle.

Expression: _____

True or false?

A 90° angle is split into x° and 25° , so
 $x + 25 = 90$.

A. True

B. False

Two equal angles of n° make a straight angle (180°).

Which equation is true?

A. $n + 180 = 2$

B. $2 + n = 180$

C. $n = 180$

D. $2n = 180$

Three equal angles of a° make a full turn (360°).

Write an equation.

Equation: _____

True or false?

An angle made of parts of y° and 30° measures $30y^\circ$.

A. True

B. False

Solve for unknown angles in diagrams and real-world problems with an equation

MAIN	Grade 4 main question	4.MD.C.7
BACKWARD 1	Solve one-step word problems with an unknown in any position	2.OA.A.1
BACKWARD 2	Find an unknown in an equation within 20	1.OA.D.8
FORWARD 1	Evaluate expressions with grouping symbols	5.OA.A.1
FORWARD 2	Write and solve one-step equations for unknown angles — neares...	6.EE.B.7

SET 57 • 4.MD.C.7

MAIN

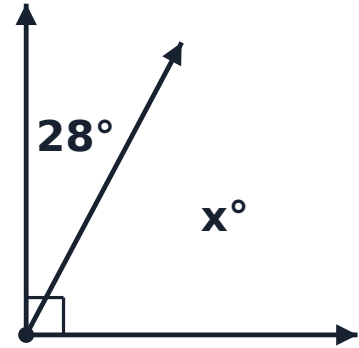
Grade 4 main question • Grade 4 • Measurement & Data • 4.MD.C.7

The whole angle is a right angle (90°).

Write an equation with a letter for the unknown part. Then solve it.

Equation: _____

Unknown angle: _____

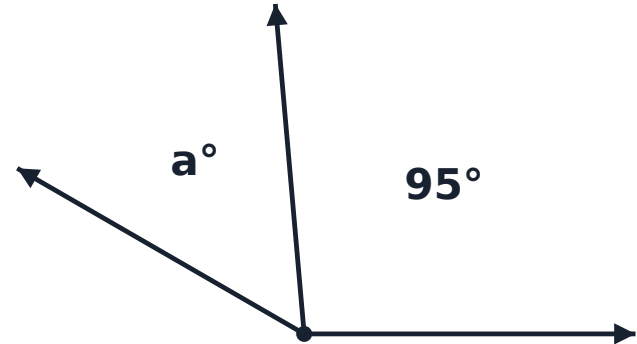


The whole angle is 150° .

Write an equation with a letter for the unknown part.
Then solve it.

Equation: _____

Unknown angle: _____



A straight angle (180°) is split into angles of 110° and m° .

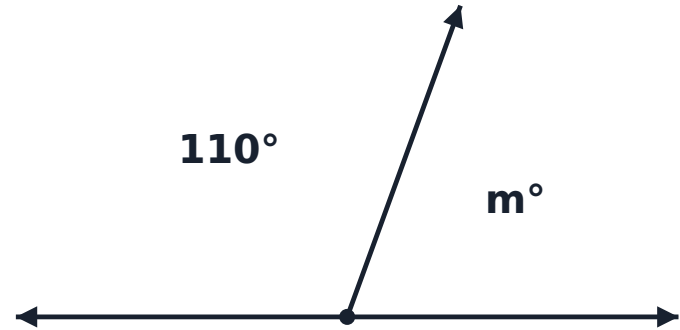
Which equation can be used to find m ?

A. $m - 110$
 $= 180$

B. $110 + m$
 $= 180$

C. $110 + m$
 $= 90$

D. $m = 110$
 $+ 180$



True or false?

A door opens 35° and then 40° more. It has turned through 75° in all.

- A. True
- B. False

A skater turns 90° and then 135° in the same direction.
How many degrees did she turn in all? How many more degrees would make a full turn (360°)?

Total turn: _____

More to a full turn: _____

SET 57 • 4.MD.C.7

BACKWARD 1

Solve one-step word problems with an unknown in any position • Grade 2 •
Operations & Algebraic Thinking • 2.OA.A.1

A box had some crayons. After 28 more crayons were put in, it had 90 crayons.

How many crayons were in the box at first?

Crayons: _____

True or false?

Kim had 45 stickers. She got some more, and now she has 80. She got 35 stickers.

A. True

B. False

Some birds were on a fence. 37 more birds came, and then there were 60 birds.

How many birds were on the fence at first?

A. 97

B. 27

C. 33

D. 23

Ty had 100 cards. He gave some away and has 64 left.
How many cards did he give away?

Cards: _____

True or false?

A shelf had some books. 19 more books were added, making 50 books. There were 41 books at first.

- A.** True
- B.** False

SET 57 • 4.MD.C.7

BACKWARD 2

Find an unknown in an equation within 20 • Grade 1 • Operations & Algebraic Thinking • 1.OA.D.8

Find the unknown number.

$$8 + ? = 15$$

Unknown: _____

True or false?

If $6 + n = 13$, then $n = 7$.

A. True

B. False

$$? - 4 = 9$$

A. 5

B. 13

C. 12

D. 4

Find the unknown number.

$$14 - ? = 6$$

Unknown: _____

True or false?

If $9 + n = 17$, then $n = 9$.

A. True

B. False

SET 57 • 4.MD.C.7

FORWARD 1

Evaluate expressions with grouping symbols • Grade 5 • Operations & Algebraic Thinking • 5.OA.A.1

Evaluate.

$$180 - (45 + 60)$$

Value: _____

True or false?

$$360 - (90 + 90 + 120) = 60$$

A. True

B. False

Evaluate.

$$90 - (12 + 33)$$

A. 45

B. 111

C. 35

D. 69

Evaluate.

$$[360 - (100 + 80)] \div 2$$

Value: _____

True or false?

$$180 - 50 + 30 = 100$$

A. True

B. False

FORWARD 2

Write and solve one-step equations for unknown angles • Grade 6 •
Expressions & Equations • 6.EE.B.7

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

Two equal angles together make 130° .

Write and solve an equation for each angle, x .

Equation: _____

$x =$ _____

True or false?

If 4 equal angles make a full turn of 360° , each angle is 90° .

A. True

B. False

Solve for x.

$$x + 72 = 180$$

A. 98

B. 108

C. 118

D. 252

Five equal angles meet at a point and make 360° .

Write and solve an equation for each angle, a .

Equation: _____

$a =$ _____

True or false?

If $x + 45 = 90$, then $x = 135$.

A. True

B. False

Identify points, lines, line segments, and rays

MAIN	Grade 4 main question	4.G.A.1
BACKWARD 1	Shapes with straight sides	1.G.A.1
BACKWARD 2	Measure the length of a segment with a ruler	2.MD.A.1
FORWARD 1	Understand the coordinate plane: perpendicular axes and the orig...	5.G.A.1
FORWARD 2	Name and plot points in all four quadrants	6.NS.C.6.c

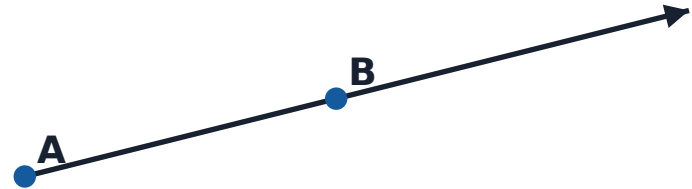
SET 58 • 4.G.A.1

MAIN

Grade 4 main question • Grade 4 • Geometry • 4.G.A.1

What is the figure called?

- A.** Point
- B.** Line segment
- C.** Line
- D.** Ray



What is the figure called?

- A.** Angle
- B.** Ray
- C.** Line segment
- D.** Line



True or false?

A line goes on without end in both directions.

A. True

B. False

What is the figure called?

- A.** Line
- B.** Line segment
- C.** Ray
- D.** Point



True or false?

A line segment has an arrow at each end.

A. True

B. False

SET 58 • 4.G.A.1

BACKWARD 1

Shapes with straight sides • Grade 1 • Geometry • 1.G.A.1

True or false?

A triangle has 3 straight sides.

A. True

B. False

Which shape has no straight sides?

- A.** Square
- B.** Triangle
- C.** Circle
- D.** Rectangle

How many straight sides does a rectangle have?

Sides: _____

True or false?

A closed shape with 4 straight sides is a circle.

A. True

B. False

A closed shape has 3 straight sides.

What is it called?

Name: _____

SET 58 • 4.G.A.1

BACKWARD 2

Measure the length of a segment with a ruler • Grade 2 • Measurement & Data
• 2.MD.A.1

A segment starts at 0 cm and ends at 8 cm on a ruler.

How long is it?

Length: _____

True or false?

You can measure the length of a line segment with a ruler.

- A.** True
- B.** False

Which tool is best for measuring the length of a segment?

- A.** A measuring cup
- B.** A clock
- C.** A scale
- D.** A ruler

A segment goes from 3 inches to 7 inches on a ruler.

How long is it?

Length: _____

True or false?

A segment from 2 cm to 6 cm on a ruler is 6 cm long.

- A.** True
- B.** False

SET 58 • 4.G.A.1

FORWARD 1

Understand the coordinate plane: perpendicular axes and the origin • Grade 5

• Geometry • 5.G.A.1

What is the point where the x-axis and the y-axis meet called?

Answer: _____

True or false?

The x-axis and the y-axis are perpendicular lines.

A. True

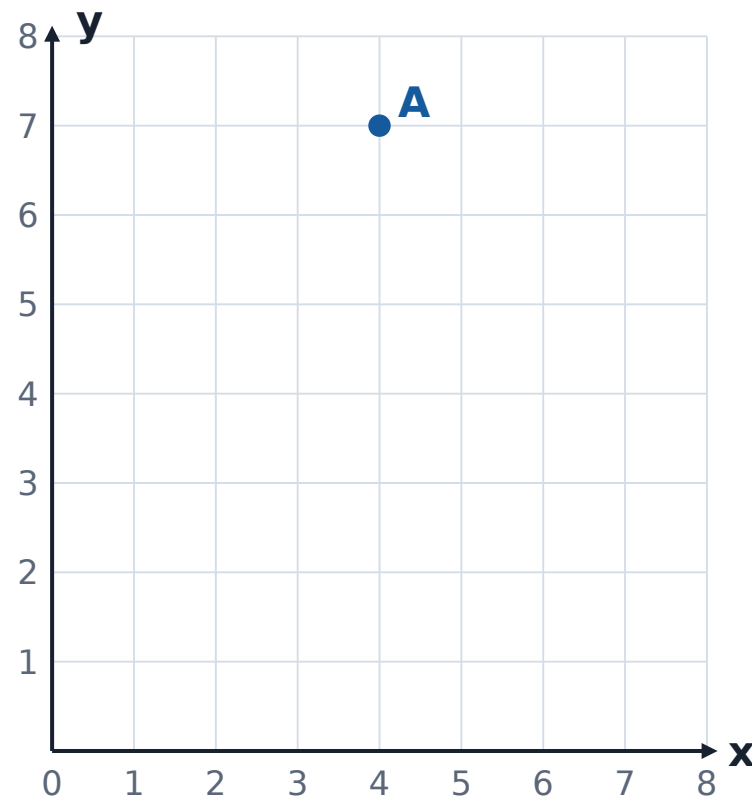
B. False

In the ordered pair (6, 2), what does the 6 tell you?

- A.** Move 6 units along the y-axis
- B.** The point is on the y-axis
- C.** Move 2 units to the right
- D.** Move 6 units along the x-axis

What are the coordinates of point A?

A = _____



True or false?

$(3, 5)$ and $(5, 3)$ name the same point.

A. True

B. False

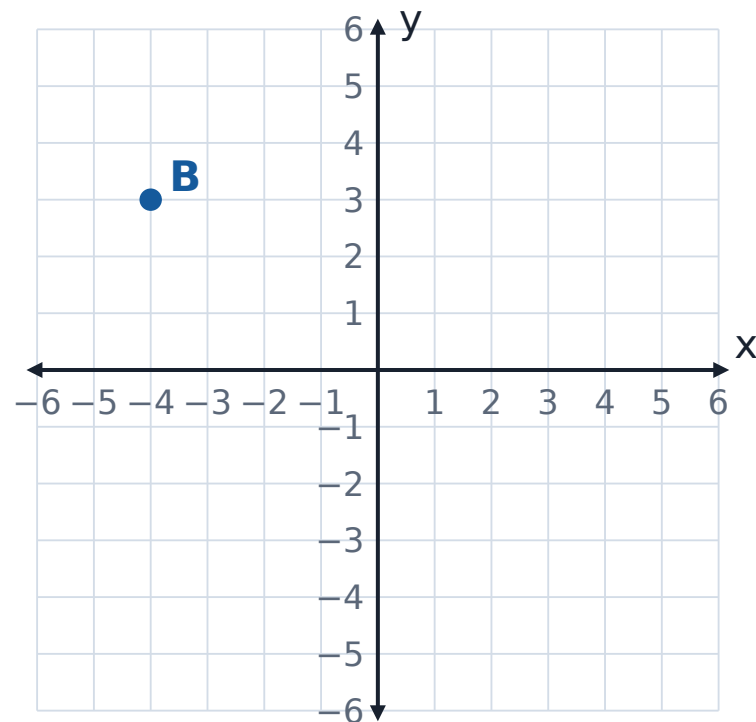
SET 58 • 4.G.A.1

FORWARD 2

Name and plot points in all four quadrants • Grade 6 • The Number System •
6.NS.C.6.c

What are the coordinates of point B?

B = _____



True or false?

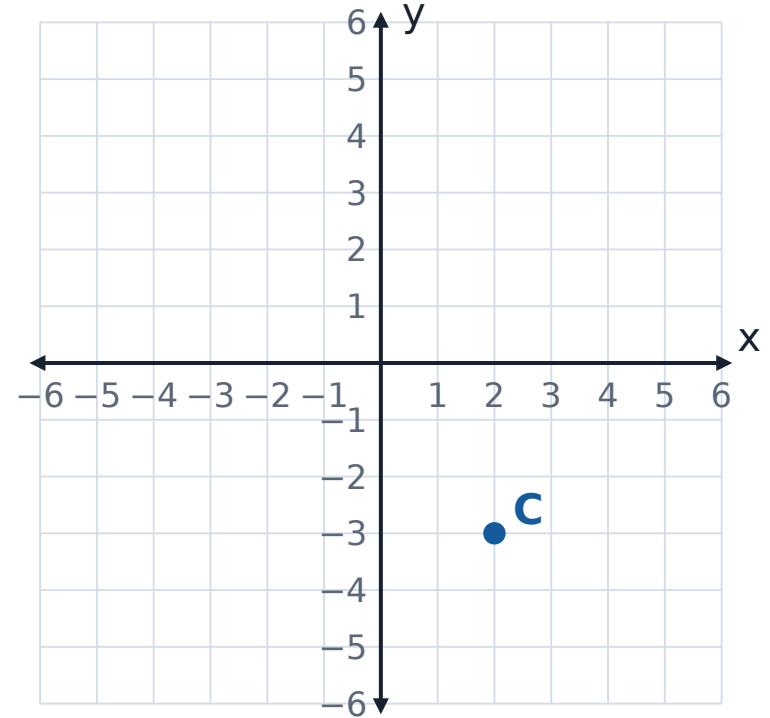
The point $(0, -5)$ is on the y -axis.

A. True

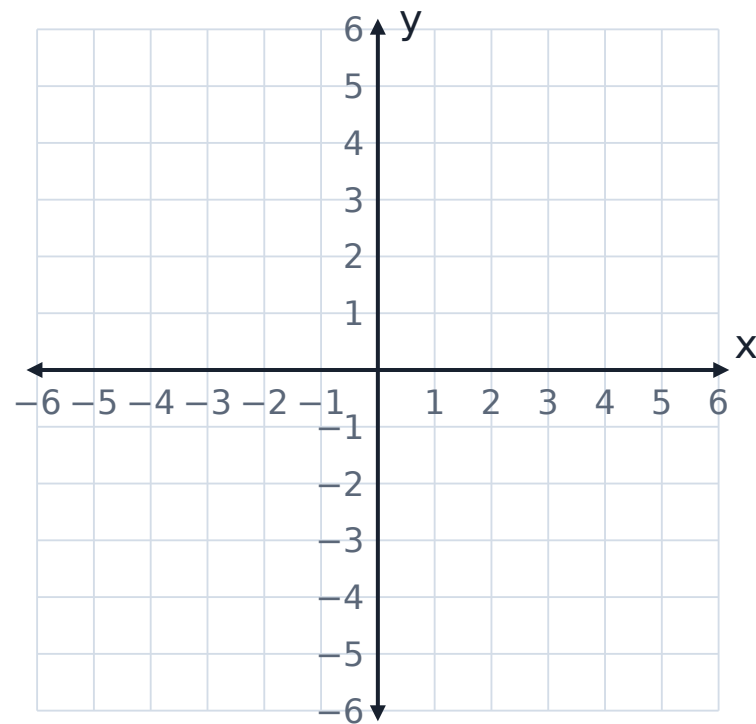
B. False

Which ordered pair names point C?

- A.** $(-3, 2)$
- B.** $(-2, 3)$
- C.** $(3, -2)$
- D.** $(2, -3)$



Plot and label the points
 $D(-2, -4)$ and $E(5, 1)$.



True or false?

The point $(-1, -1)$ is in Quadrant II.

A. True

B. False

Identify right, acute, and obtuse angles

MAIN	Grade 4 main question	4.G.A.1
BACKWARD 1	Shapes with a given number of angles	2.G.A.1
BACKWARD 2	Rectangles and squares have square corners	3.G.A.1
FORWARD 1	Classify triangles by their angles	5.G.B.4
FORWARD 2	Use the right angle of a right triangle to find its area — nearest re...	6.G.A.1

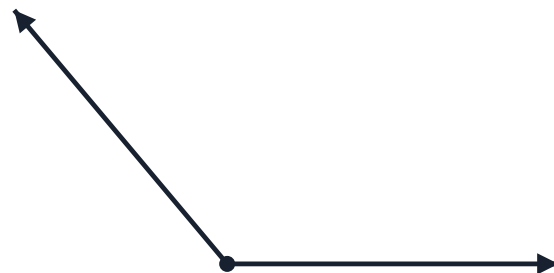
SET 59 • 4.G.A.1

MAIN

Grade 4 main question • Grade 4 • Geometry • 4.G.A.1

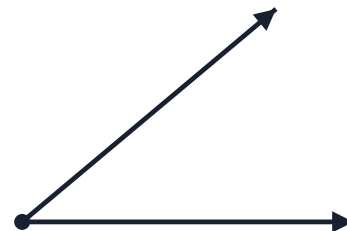
What kind of angle is shown?

- A.** Acute
- B.** Right
- C.** Obtuse
- D.** Straight



What kind of angle is shown?

- A.** Acute
- B.** Obtuse
- C.** Right
- D.** Straight



True or false?

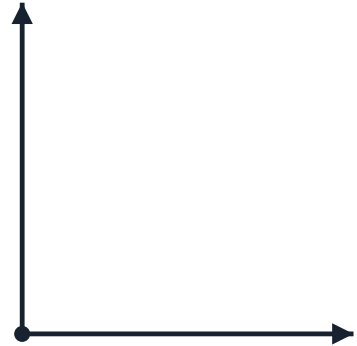
A right angle measures exactly 180° .

A. True

B. False

What kind of angle is shown?

- A.** Straight
- B.** Obtuse
- C.** Acute
- D.** Right



Classify each angle as acute, right, or obtuse.

15° 90° 165°

Answers: _____

SET 59 • 4.G.A.1

BACKWARD 1

Shapes with a given number of angles • Grade 2 • Geometry • 2.G.A.1

How many angles does a hexagon have?

Angles: _____

True or false?

A quadrilateral has 4 angles.

A. True

B. False

Which shape has 3 angles?

- A.** Triangle
- B.** Pentagon
- C.** Square
- D.** Hexagon

How many angles does a rectangle have?

Angles: _____

True or false?

A pentagon has 6 angles.

A. True

B. False

SET 59 • 4.G.A.1

BACKWARD 2

Rectangles and squares have square corners • Grade 3 • Geometry • 3.G.A.1

True or false?

Every rectangle has 4 square corners.

A. True

B. False

Which shape always has 4 square corners?

- A.** Trapezoid
- B.** Triangle
- C.** Square
- D.** Rhombus

How many square corners does a square have?

Square corners: _____

True or false?

Every quadrilateral has 4 square corners.

A. True

B. False

Name a quadrilateral that has 4 square corners but is not a square.

Shape: _____

SET 59 • 4.G.A.1

FORWARD 1

Classify triangles by their angles • Grade 5 • Geometry • 5.G.B.4

A triangle has angles of 100° , 40° , and 40° .
Classify it by its angles.

Classification: _____

True or false?

A right triangle has exactly one right angle.

A. True

B. False

Which kind of triangle has three acute angles?

- A.** An obtuse triangle
- B.** An acute triangle
- C.** A straight triangle
- D.** A right triangle

A triangle has angles of 55° , 35° , and 90° .
Classify it by its angles.

Classification: _____

True or false?

An acute triangle can have one obtuse angle.

A. True

B. False

SET 59 • 4.G.A.1

FORWARD 2

Use the right angle of a right triangle to find its area • Grade 6 • Geometry • 6.G.A.1

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

A right triangle has legs that are 6 cm and 10 cm long.
What is its area?

Area: _____

True or false?

In a right triangle, the two sides that form the right angle can be used as the base and the height.

- A.** True
- B.** False

An 8-by-5 rectangle is cut along a diagonal into two triangles.

What is the area of each triangle?

- A.** 40 square units
- B.** 13 square units
- C.** 20 square units
- D.** 10 square units

A triangle has a base of 12 m and a height of 7 m.
What is its area?

Area: _____

True or false?

An obtuse triangle does not have a height.

A. True

B. False

Identify perpendicular and parallel lines

MAIN	Grade 4 main question	4.G.A.1
BACKWARD 2	Quadrilaterals share attributes	3.G.A.1
BACKWARD 3	Square corners and sides that run the same way in a rectangle	3.G.A.1
FORWARD 1	Attributes of a category belong to all its subcategories	5.G.B.3
FORWARD 2	Use coordinates to find parallel and perpendicular sides of polygo...	6.G.A.3

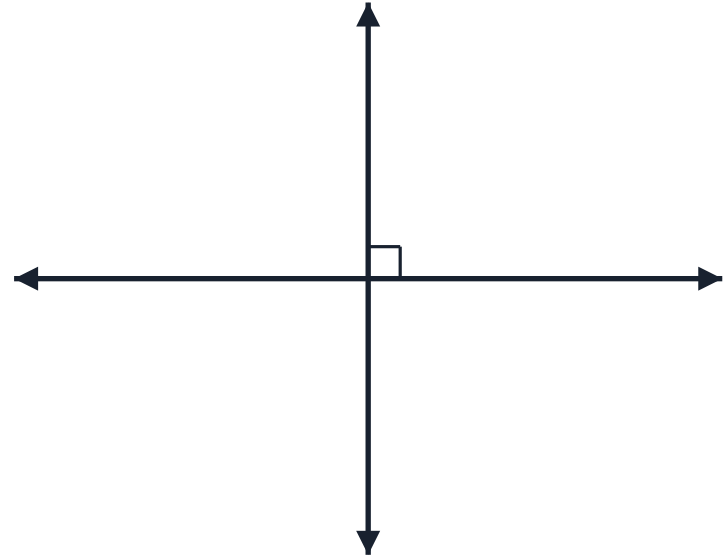
SET 60 • 4.G.A.1

MAIN

Grade 4 main question • Grade 4 • Geometry • 4.G.A.1

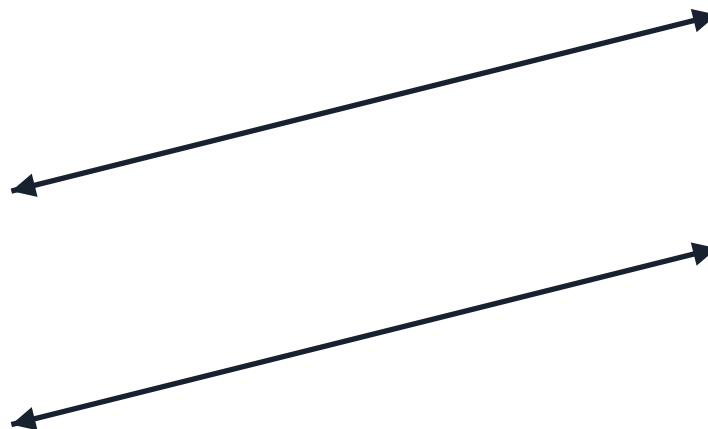
How are the two lines related?

- A.** Parallel
- B.** Perpendicular
- C.** Neither
- D.** They are the same line



How are the two lines related?

- A.** Parallel
- B.** Perpendicular
- C.** Neither
- D.** They are the same line



True or false?

Parallel lines never meet.

A. True

B. False

Lines a and b meet to form a right angle.
Are they parallel or perpendicular?

Answer: _____

True or false?

Perpendicular lines meet to form four obtuse angles.

A. True

B. False

SET 60 • 4.G.A.1

BACKWARD 2

Quadrilaterals share attributes • Grade 3 • Geometry • 3.G.A.1

True or false?

A rhombus, a rectangle, and a square are all quadrilaterals.

A. True

B. False

Which shape is a quadrilateral?

A. Pentagon

B. Triangle

C. Circle

D. Rhombus

How many sides does every quadrilateral have?

Sides: _____

True or false?

A triangle is a quadrilateral.

A. True

B. False

Name a quadrilateral that is not a rectangle.

Shape: _____

SET 60 • 4.G.A.1

BACKWARD 3

Square corners and sides that run the same way in a rectangle • Grade 3 •
Geometry • 3.G.A.1

True or false?

In a rectangle, the two sides that meet at a corner make a square corner.

A. True

B. False

In a rectangle, which sides run the same way and never meet?

- A.** All four sides
- B.** The two opposite sides
- C.** Two sides that meet at a corner
- D.** No sides

How many pairs of opposite sides that run the same way does a rectangle have?

Pairs: _____

True or false?

Two sides of a rectangle that meet at a corner run the same way.

A. True

B. False

Which shape has 4 square corners and opposite sides that run the same way?

- A.** Circle
- B.** Rectangle
- C.** Triangle
- D.** Hexagon

SET 60 • 4.G.A.1

FORWARD 1

Attributes of a category belong to all its subcategories • Grade 5 • Geometry •
5.G.B.3

True or false?

All parallelograms have two pairs of parallel sides.
A rectangle is a parallelogram, so every rectangle has two pairs of parallel sides.

A. True

B. False

Which shape must have two pairs of parallel sides?

- A.** Triangle
- B.** Kite
- C.** Trapezoid
- D.** Rhombus

Do the sides of every square meet at right angles?
Explain.

Answer: _____

Explanation: _____

True or false?

Every quadrilateral has at least one pair of parallel sides.

- A. True
- B. False

Which shape has exactly one pair of parallel sides?

- A.** A square
- B.** A trapezoid with only one pair of parallel sides
- C.** A rectangle
- D.** A rhombus

SET 60 • 4.G.A.1

FORWARD 2

Use coordinates to find parallel and perpendicular sides of polygons • Grade 6
• Geometry • 6.G.A.3

True or false?

The segment from $(1, 2)$ to $(5, 2)$ is parallel to the segment from $(1, -3)$ to $(5, -3)$.

A. True

B. False

What is the length of the side from $(-3, -1)$ to $(-3, 4)$?

Length: _____

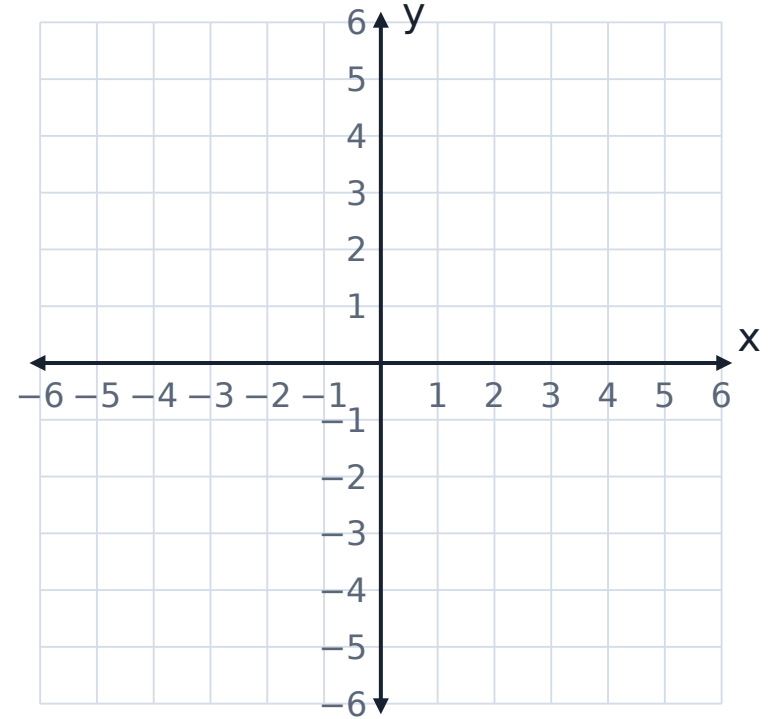
Which segment is perpendicular to the segment from $(0, 0)$ to $(6, 0)$?

- A.** From $(0, 1)$ to $(6, 1)$
- B.** From $(2, 0)$ to $(2, 5)$
- C.** From $(1, 1)$ to $(4, 4)$
- D.** From $(0, 0)$ to $(6, 3)$

Plot $(-2, 1)$, $(4, 1)$, $(4, -3)$,
and $(-2, -3)$. Connect them
in order.

How many pairs of parallel
sides does the polygon have?

Pairs: _____



True or false?

The segment from $(0, 1)$ to $(3, 1)$ is perpendicular to the segment from $(0, 4)$ to $(3, 4)$.

- A. True
- B. False

Draw points, lines, line segments, and rays

MAIN	Grade 4 main question	4.G.A.1
BACKWARD 1	Draw shapes with straight sides	2.G.A.1
BACKWARD 2	Measure the length of a drawn segment	2.MD.A.1
FORWARD 1	Locate and draw points and segments on a coordinate grid	5.G.A.1
FORWARD 2	Draw points and segments in all four quadrants	6.NS.C.6.c

SET 70 • 4.G.A.1

MAIN

Grade 4 main question • Grade 4 • Geometry • 4.G.A.1

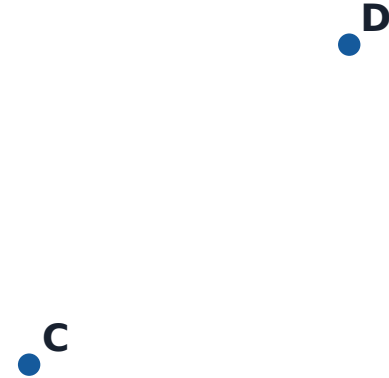
Draw ray PQ. It starts at point P and goes through point Q.



Draw line segment AB.



Draw line CD through points
C and D.



Draw a point that is not on line m , and label it E .



Draw ray ST. It starts at point S and goes through point T.

How is a ray different from a line segment?



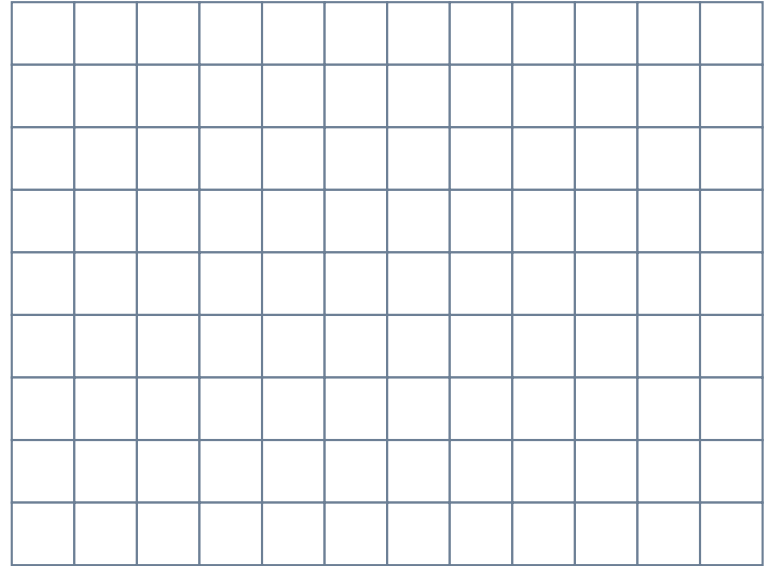
Difference: _____

SET 70 • 4.G.A.1

BACKWARD 1

Draw shapes with straight sides • Grade 2 • Geometry • 2.G.A.1

Draw a closed shape with 4 straight sides.



True or false?

A closed shape drawn with 3 straight sides that meet at 3 corners is a triangle.

A. True

B. False

Which shape has only straight sides?

- A.** Pentagon
- B.** Oval
- C.** Circle
- D.** Half-circle

How many straight sides do you draw to make a hexagon?

Sides: _____

True or false?

A circle is drawn with straight sides.

A. True

B. False

SET 70 • 4.G.A.1

BACKWARD 2

Measure the length of a drawn segment • Grade 2 • Measurement & Data •
2.MD.A.1

A segment starts at 0 and ends at 5 on a centimeter ruler.

How long is it?

Length: _____

True or false?

To draw a segment 6 cm long, you can start at 0 on the ruler and stop at 6.

- A.** True
- B.** False

A segment is drawn from 2 to 9 on an inch ruler.

How long is it?

- A.** 9 inches
- B.** 11 inches
- C.** 7 inches
- D.** 2 inches

Ben draws a segment from 0 to 4 on an inch ruler.
How long is his segment?

Length: _____

True or false?

A segment drawn from 1 cm to 5 cm on a ruler is 5 cm long.

A. True

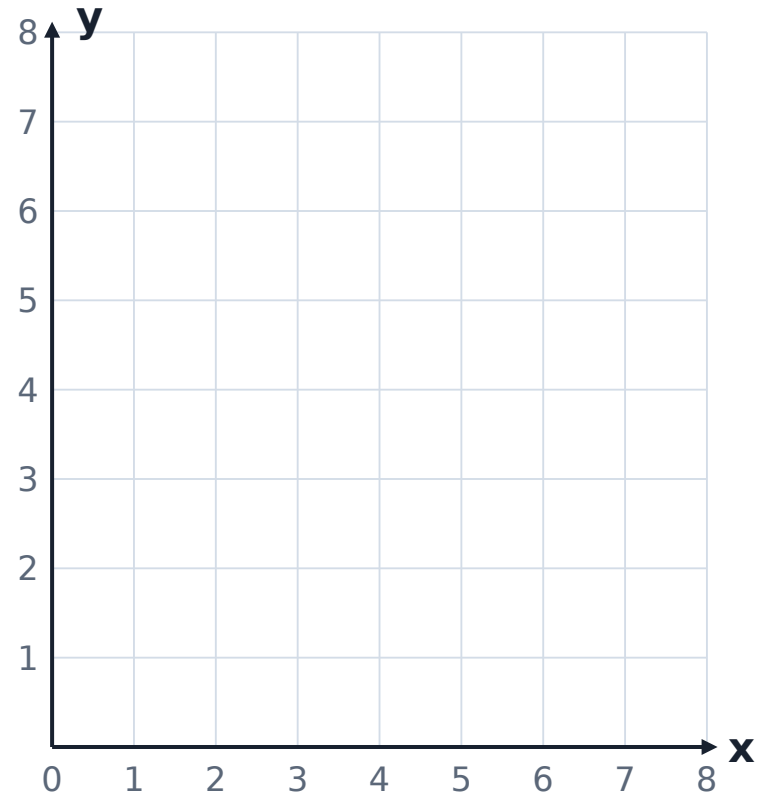
B. False

SET 70 • 4.G.A.1

FORWARD 1

Locate and draw points and segments on a coordinate grid • Grade 5 •
Geometry • 5.G.A.1

Plot the point $(3, 5)$ and label it P.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q1

Grade 5 • Geometry • 5.G.A.1 • Set 70 • FORWARD 1 • Q1/5

True or false?

To locate $(4, 2)$, start at the origin, move 4 units along the x-axis, then move 2 units up.

A. True

B. False

Which point is on the y-axis?

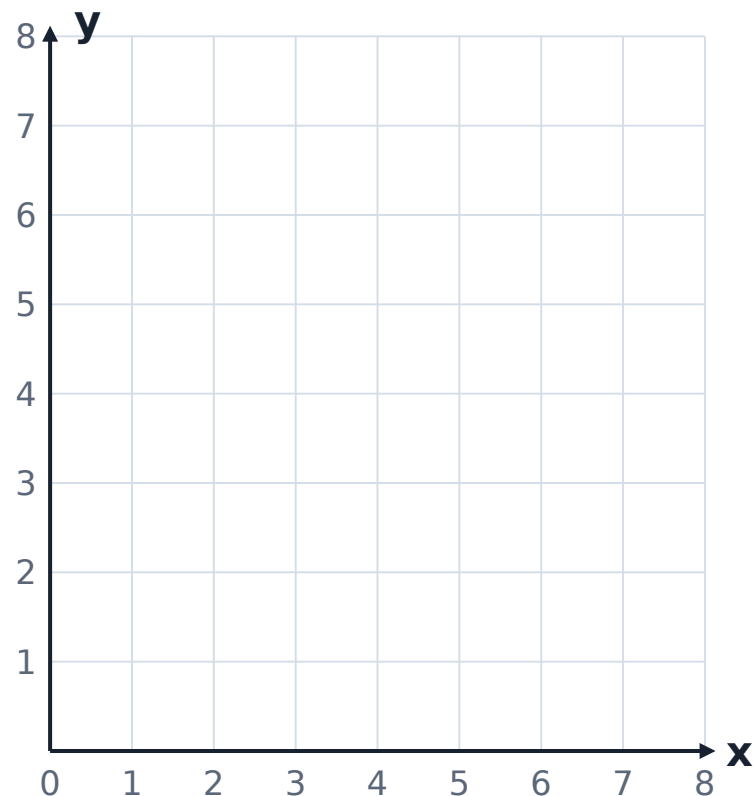
A. (1, 6)

B. (6, 0)

C. (6, 6)

D. (0, 6)

Draw the line segment from $(1, 1)$ to $(6, 1)$.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S70-F1-Q4

Grade 5 • Geometry • 5.G.A.1 • Set 70 • FORWARD 1 • Q4/5

True or false?

The origin is the point (1, 1).

A. True

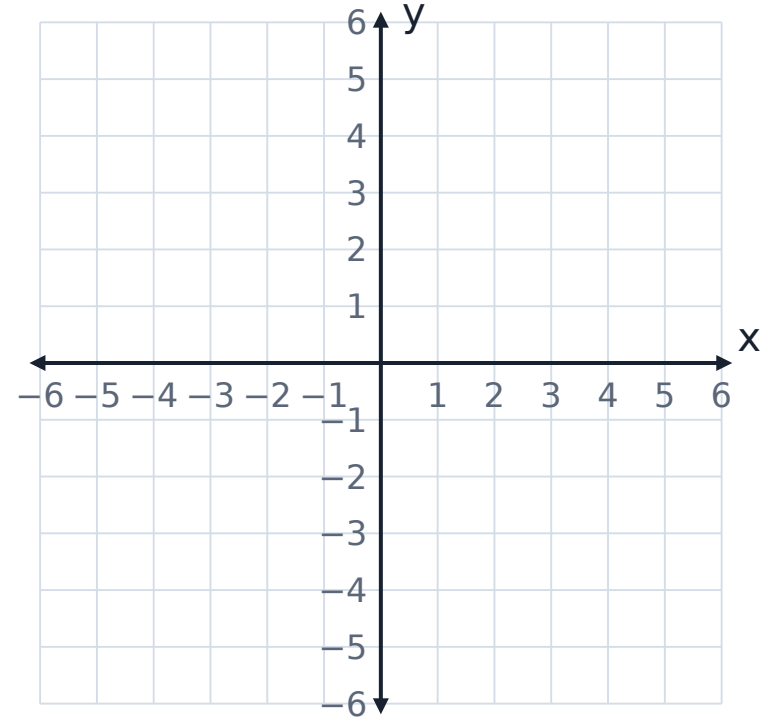
B. False

SET 70 • 4.G.A.1

FORWARD 2

Draw points and segments in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

Plot and label the points
 $A(-3, 2)$ and $B(4, -1)$.



True or false?

The point $(-5, 0)$ is on the x-axis.

A. True

B. False

Which point is in Quadrant III?

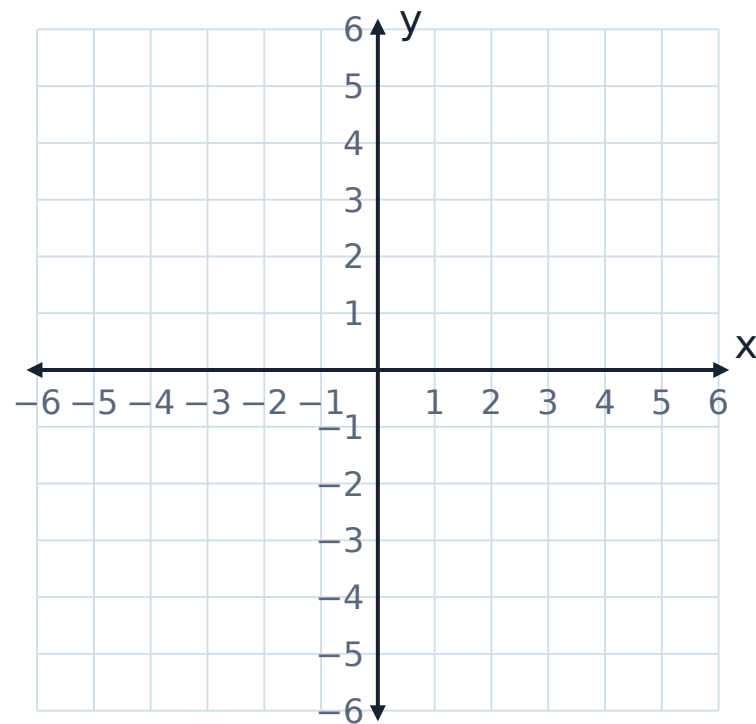
A. $(-2, 4)$

B. $(2, -4)$

C. $(-2, -4)$

D. $(2, 4)$

Draw the segment from $(-4, -3)$ to $(2, -3)$.



True or false?

The point $(3, -2)$ is in Quadrant II.

A. True

B. False

Draw right, acute, and obtuse angles

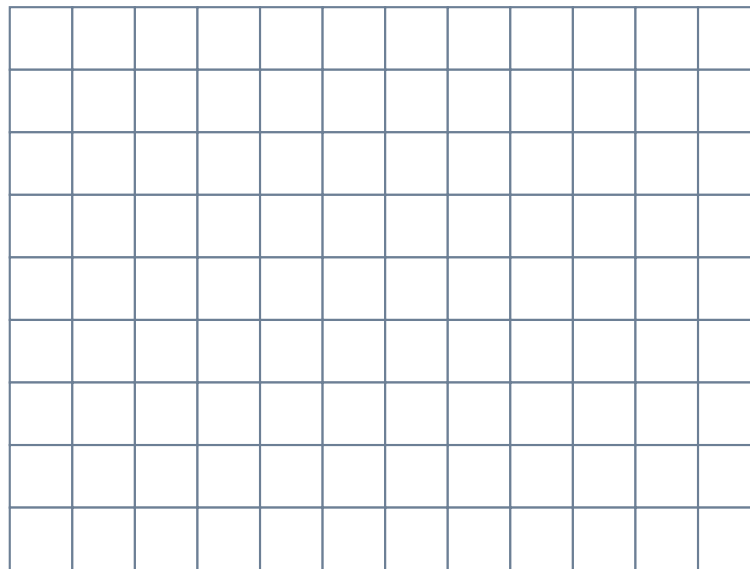
MAIN	Grade 4 main question	4.G.A.1
BACKWARD 1	Square corners	3.G.A.1
BACKWARD 2	Draw shapes with a given number of angles	2.G.A.1
FORWARD 1	Draw and classify triangles by their angles	5.G.B.4
FORWARD 2	Draw polygons with right angles in the coordinate plane — neares...	6.G.A.3

SET 71 • 4.G.A.1

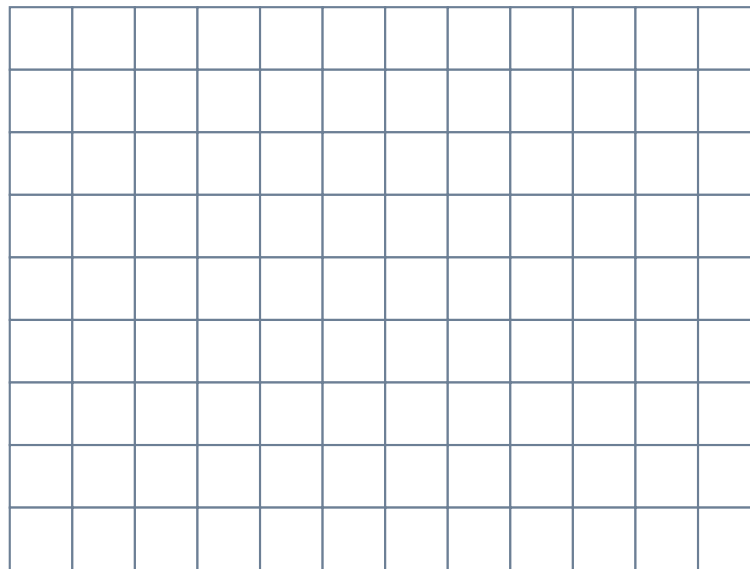
MAIN

Grade 4 main question • Grade 4 • Geometry • 4.G.A.1

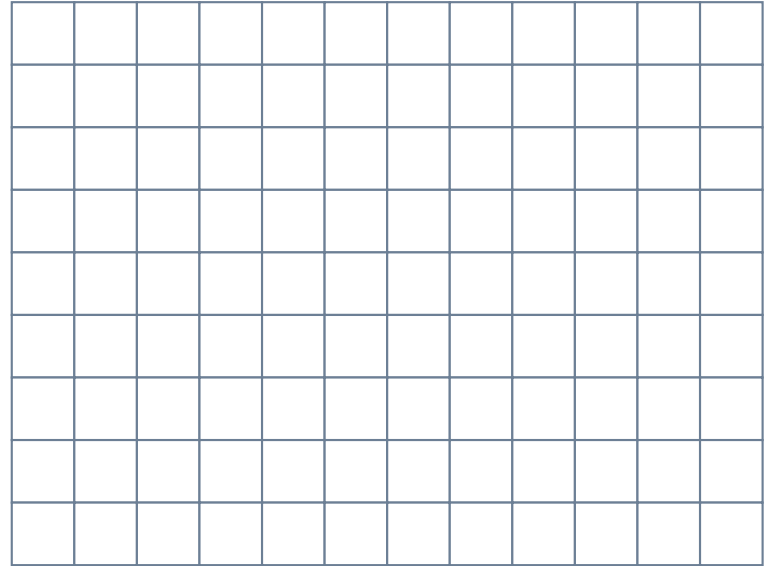
Draw an obtuse angle.



Draw an acute angle.



Draw a right angle. Mark it with a right-angle mark.



Point A is the vertex. Draw ray AB and ray AC so that angle BAC is obtuse.

Is angle BAC greater or less than a right angle?



Answer: _____

Draw an acute angle with its vertex at point V.

How do you know your angle is acute?



Explanation: _____

SET 71 • 4.G.A.1

BACKWARD 1

Square corners • Grade 3 • Geometry • 3.G.A.1

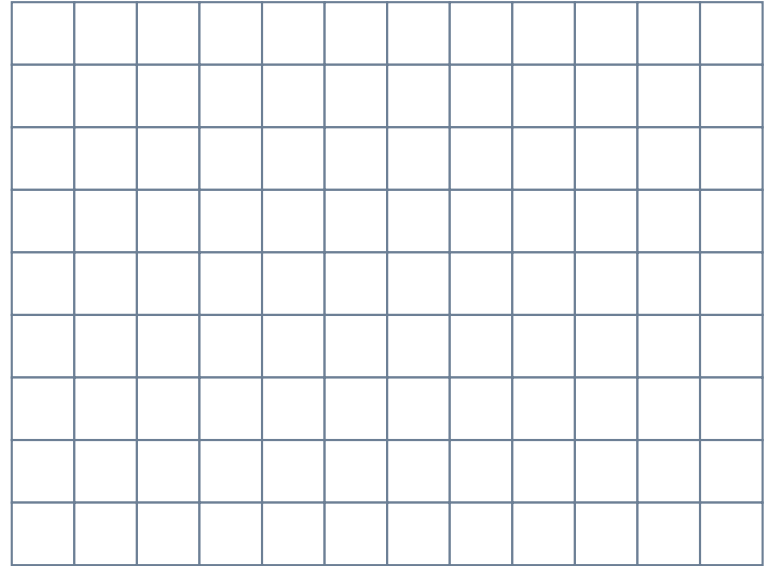
True or false?

The corner of a sheet of paper is a square corner.

A. True

B. False

Draw a shape that has 4 square corners.



Which shape always has 4 square corners?

- A.** Rectangle
- B.** Pentagon
- C.** Trapezoid
- D.** Rhombus

A rectangle is cut into 2 smaller rectangles by one straight cut from one side to the opposite side.

How many square corners does each smaller rectangle have?

Square corners: _____

True or false?

Every triangle has a square corner.

A. True

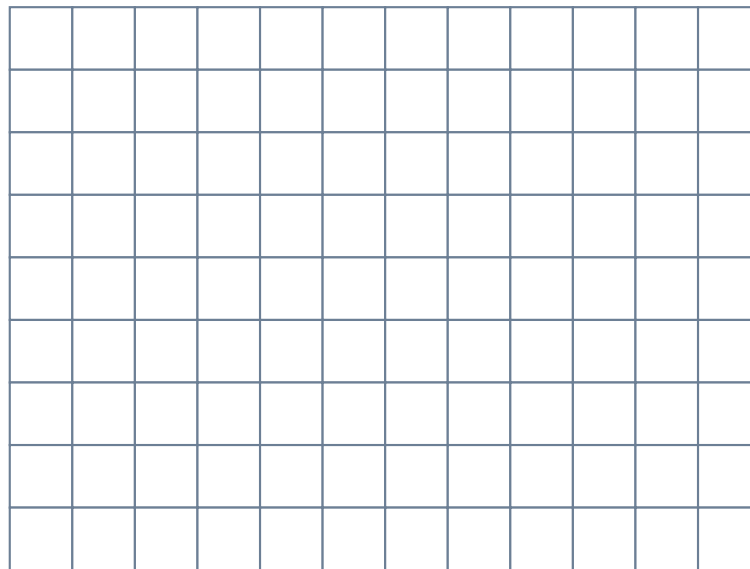
B. False

SET 71 • 4.G.A.1

BACKWARD 2

Draw shapes with a given number of angles • Grade 2 • Geometry • 2.G.A.1

Draw a shape with 3 angles.



True or false?

A closed shape with 5 angles is a pentagon.

A. True

B. False

How many angles does a hexagon have?

A. 8

B. 4

C. 5

D. 6

How many angles do you draw when you draw a quadrilateral?

Angles: _____

True or false?

A closed shape with 4 angles has 3 sides.

A. True

B. False

SET 71 • 4.G.A.1

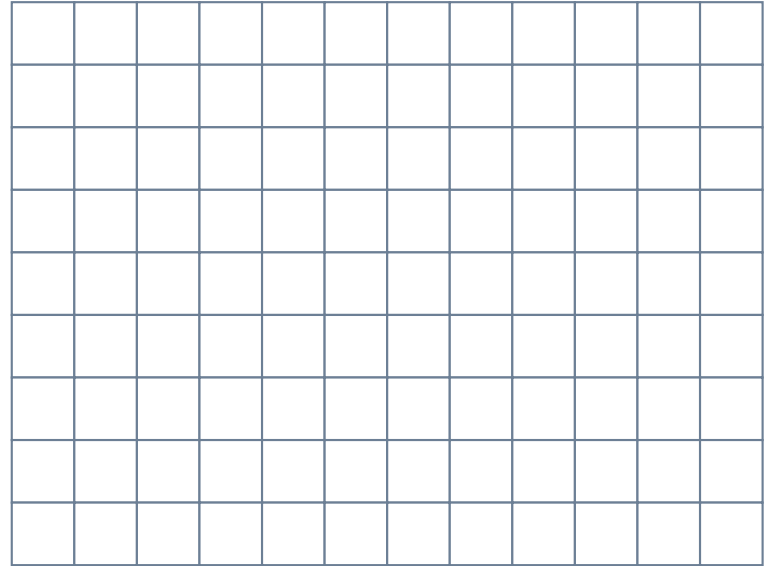
FORWARD 1

Draw and classify triangles by their angles • Grade 5 • Geometry • 5.G.B.4

Draw a triangle with one obtuse angle.

What kind of triangle is it?

Kind of triangle: _____



True or false?

A triangle can have one right angle and two acute angles.

A. True

B. False

A triangle has angles of 30° , 60° , and 90° .

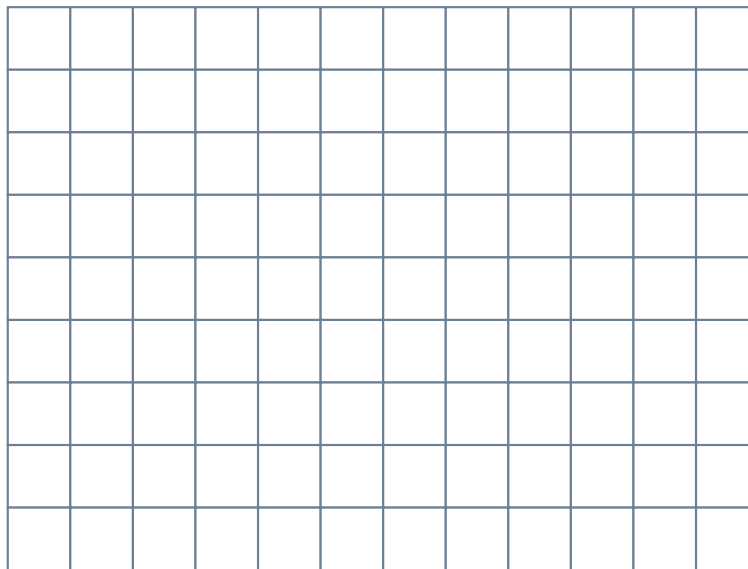
What kind of triangle is it?

- A.** Acute
- B.** Right
- C.** Equilateral
- D.** Obtuse

Draw a triangle with three acute angles.

What kind of triangle is it?

Kind of triangle: _____



True or false?

A triangle can have two obtuse angles.

A. True

B. False

FORWARD 2

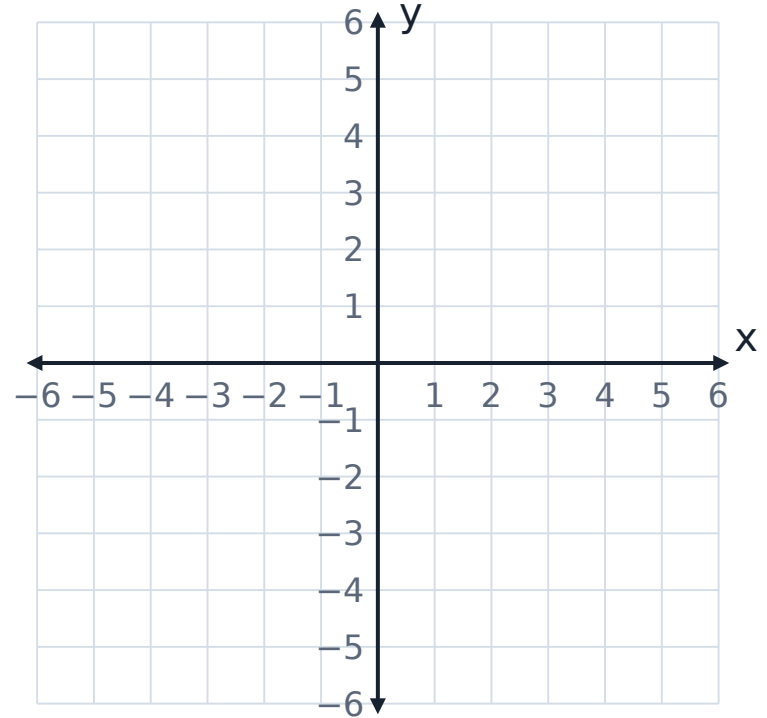
Draw polygons with right angles in the coordinate plane • Grade 6 • Geometry
• 6.G.A.3

► No Grade 6 standard directly continues this skill. This branch uses the nearest related Grade 6 standard.

Plot $(-3, -2)$, $(-3, 3)$, and $(2, -2)$, and connect them.

What kind of angle is at $(-3, -2)$?

Kind of angle: _____



True or false?

The segments from $(0, 0)$ to $(0, 4)$ and from $(0, 0)$ to $(5, 0)$ form a right angle.

- A. True
- B. False

The segment from $(1, 1)$ to $(1, 5)$ is drawn.

Which point makes a right angle at $(1, 1)$ when it is joined to $(1, 1)$?

A. $(4, 4)$

B. $(4, 5)$

C. $(4, 1)$

D. $(0, 5)$

A rectangle has vertices $(-2, 1)$, $(3, 1)$, and $(3, -4)$.

What is the fourth vertex?

Vertex: _____

True or false?

The segments from $(0, 0)$ to $(3, 3)$ and from $(0, 0)$ to $(4, 0)$ form a right angle.

- A. True
- B. False

Draw perpendicular and parallel lines

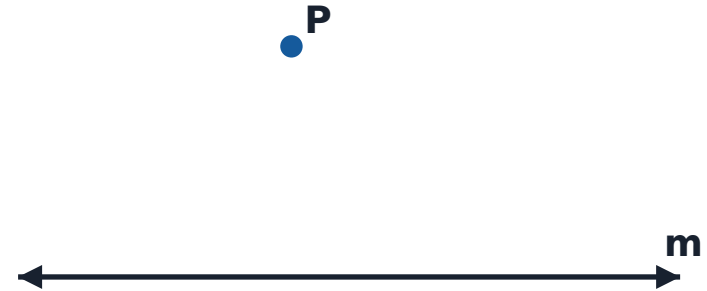
MAIN	Grade 4 main question	4.G.A.1
BACKWARD 1	Draw shapes with square corners and opposite sides that run the...	3.G.A.1
BACKWARD 2	Draw lines to split a rectangle into rows and columns of squares	2.G.A.2
FORWARD 1	Draw quadrilaterals with given parallel and perpendicular sides	5.G.B.3
FORWARD 2	Draw parallel and perpendicular segments in the coordinate plane	6.G.A.3

SET 72 • 4.G.A.1

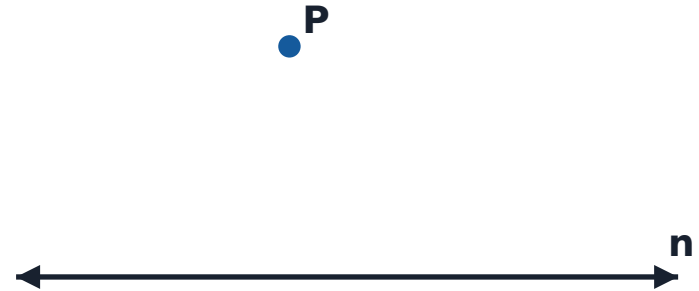
MAIN

Grade 4 main question • Grade 4 • Geometry • 4.G.A.1

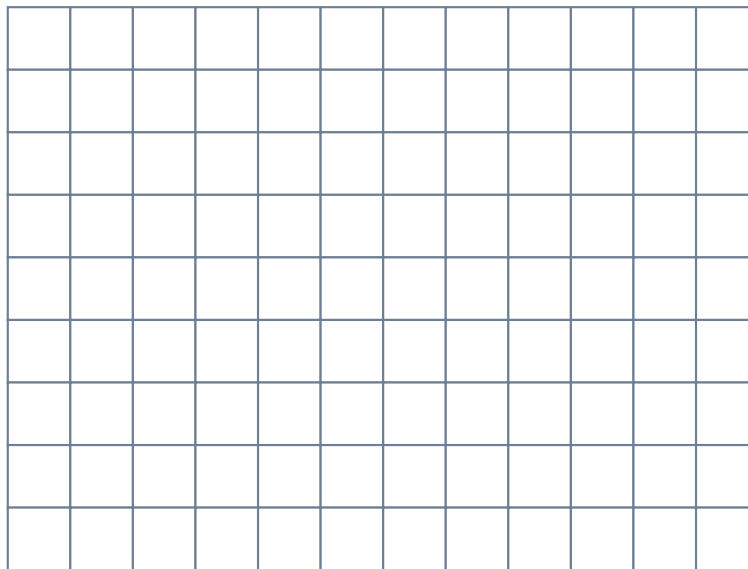
Draw a line through point P
that is parallel to line m.



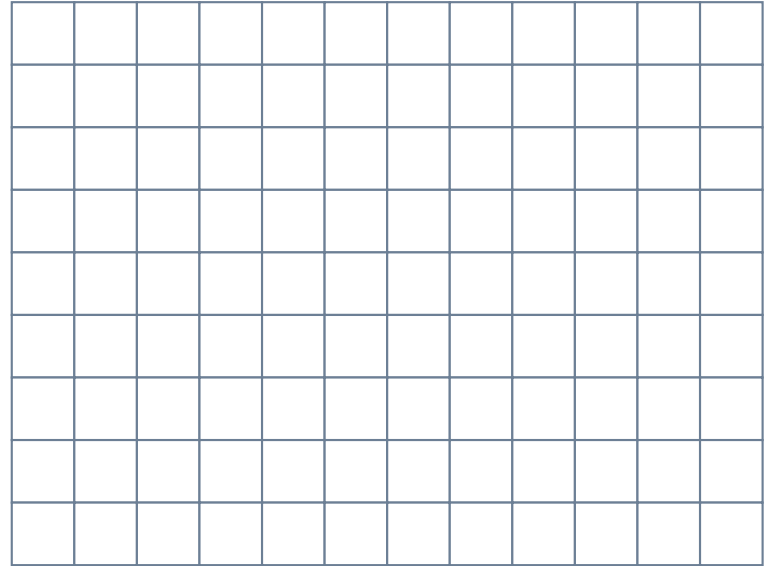
Draw a line through point P
that is perpendicular to line
n.



Draw two lines that are perpendicular to each other. Mark one of the right angles.

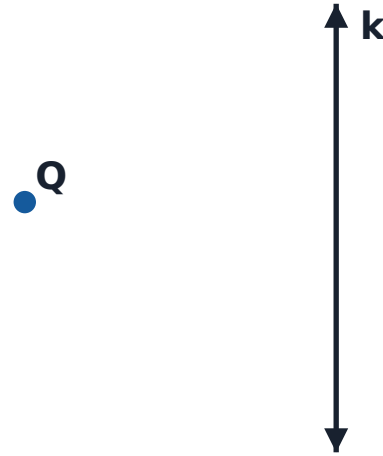


Draw two parallel line segments that are each 4 units long.



Draw a line through point Q
that is perpendicular to line k.

How many right angles do
the two lines make where
they cross?



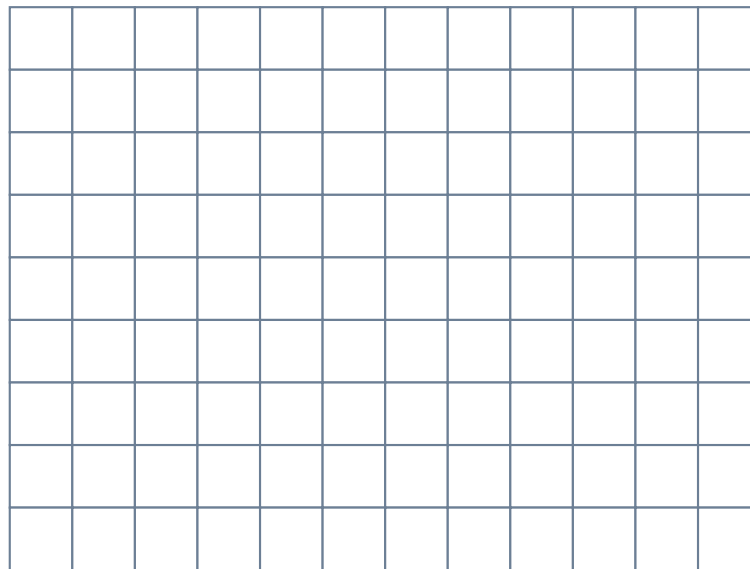
Right angles: _____

SET 72 • 4.G.A.1

BACKWARD 1

Draw shapes with square corners and opposite sides that run the same way •
Grade 3 • Geometry • 3.G.A.1

Draw a rectangle on the grid.



True or false?

Every rectangle has 2 pairs of opposite sides that run the same way.

A. True

B. False

Ann draws a shape with 4 square corners and 4 equal sides.

What shape did she draw?

A. Triangle

B. Trapezoid

C. Hexagon

D. Square

How many square corners must you draw to make a rectangle?

Square corners: _____

True or false?

A shape drawn with 4 square corners can have 5 sides.

A. True

B. False

SET 72 • 4.G.A.1

BACKWARD 2

Draw lines to split a rectangle into rows and columns of squares • Grade 2 •
Geometry • 2.G.A.2

Draw lines to split the rectangle into 2 rows of 3 same-size squares.



True or false?

To split a rectangle into 3 rows, you can draw 2 lines across it that run the same way.

- A.** True
- B.** False

A rectangle is split into 2 rows of 4 same-size squares.

How many squares are there?

A. 4

B. 8

C. 2

D. 6

How many lines across do you draw to split a rectangle into 4 rows?

Lines: _____

True or false?

A rectangle split into 2 rows and 2 columns of squares has 2 squares.

- A. True
- B. False

SET 72 • 4.G.A.1

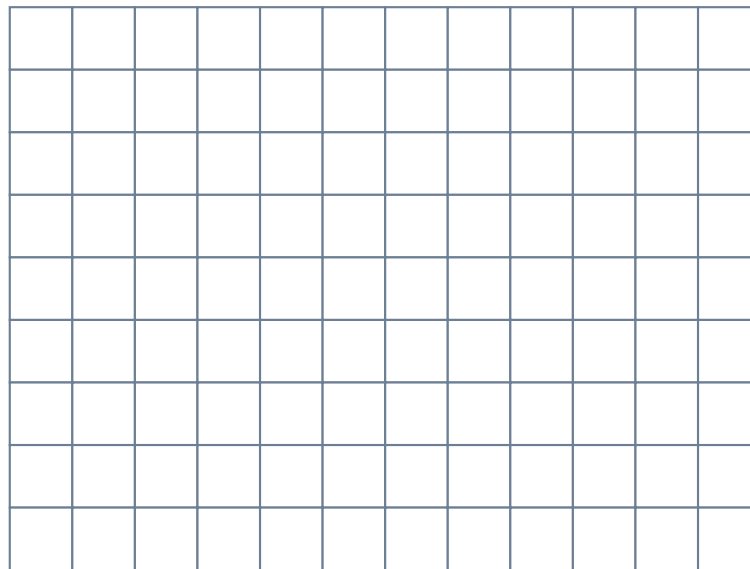
FORWARD 1

Draw quadrilaterals with given parallel and perpendicular sides • Grade 5 •
Geometry • 5.G.B.3

Draw a quadrilateral with two pairs of parallel sides and no right angles.

What is it called?

Name: _____



True or false?

Every square has two pairs of parallel sides and four right angles.

A. True

B. False

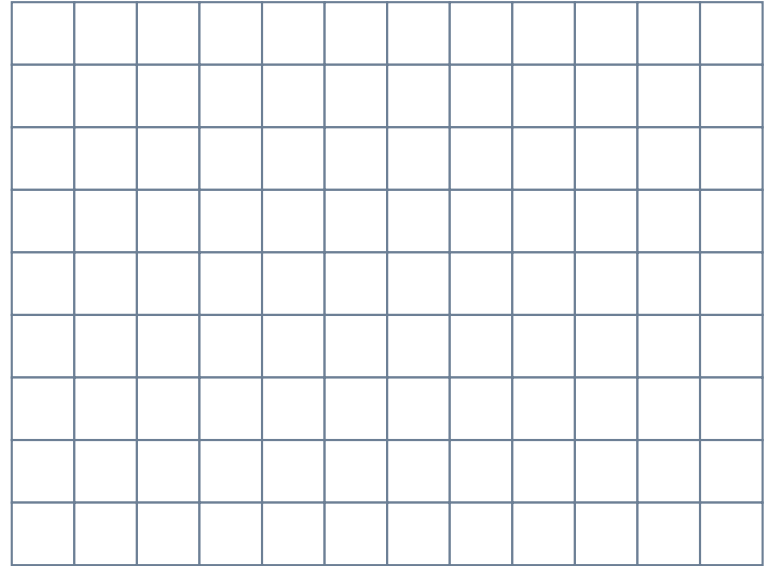
Which shape must have perpendicular sides?

- A.** Rectangle
- B.** Trapezoid
- C.** Parallelogram
- D.** Rhombus

Draw a trapezoid with exactly one pair of parallel sides and two right angles.

How many pairs of perpendicular sides does it have?

Pairs: _____



True or false?

Every parallelogram has perpendicular sides.

A. True

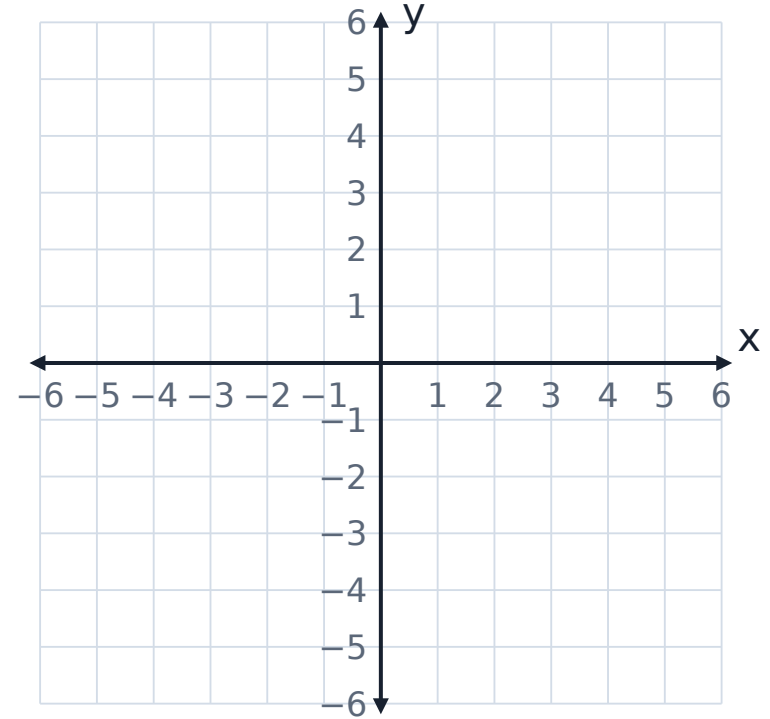
B. False

SET 72 • 4.G.A.1

FORWARD 2

Draw parallel and perpendicular segments in the coordinate plane • Grade 6 •
Geometry • 6.G.A.3

Draw the segment from $(-3, 2)$ to $(3, 2)$. Then draw a segment parallel to it that starts at $(-3, -2)$.



True or false?

The segment from $(2, -4)$ to $(2, 3)$ is perpendicular to the x-axis.

- A. True
- B. False

Which segment is parallel to the segment from $(-4, 1)$ to $(4, 1)$?

- A.** From $(-4, 1)$ to $(4, 5)$
- B.** From $(1, -4)$ to $(1, 4)$
- C.** From $(0, 0)$ to $(4, 4)$
- D.** From $(-4, -3)$ to $(4, -3)$

A segment goes from $(-1, -2)$ to $(-1, 5)$. A segment perpendicular to it starts at $(-1, 0)$.

Give one possible other endpoint.

Endpoint: _____

True or false?

The segment from $(0, 2)$ to $(5, 2)$ is perpendicular to the segment from $(0, -1)$ to $(5, -1)$.

- A. True
- B. False

Identify segments, angles, and perpendicular and parallel sides in two-dimensional figures

MAIN	Grade 4 main question	4.G.A.1
BACKWARD 1	Count sides and angles of shapes	2.G.A.1
BACKWARD 2	Find square corners in shapes	3.G.A.1
FORWARD 1	Use the attributes of a category to find sides and angles in figures	5.G.B.3
FORWARD 2	Identify parallel and perpendicular sides of polygons in the coordi...	6.G.A.3

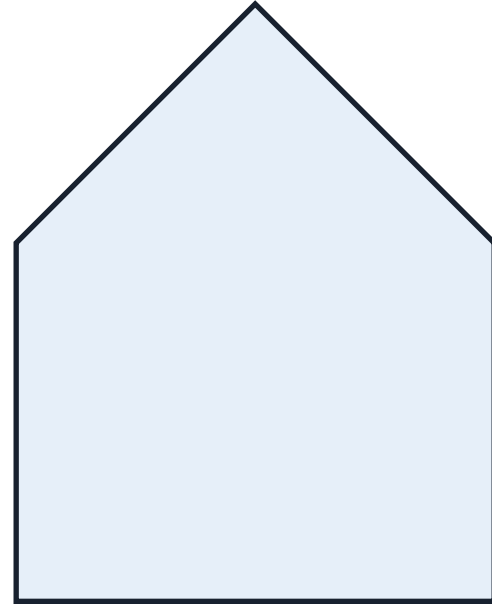
SET 73 • 4.G.A.1

MAIN

Grade 4 main question • Grade 4 • Geometry • 4.G.A.1

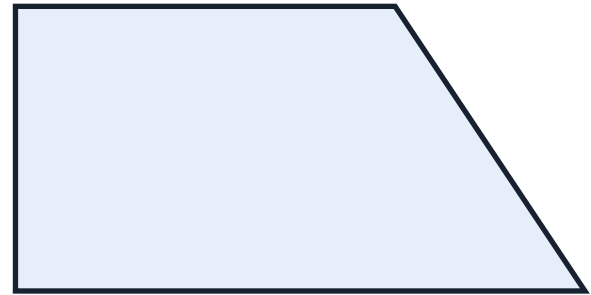
How many right angles does the figure have?

- A.** 1
- B.** 3
- C.** 2
- D.** 5



How many pairs of parallel sides does the figure have?

Pairs: _____



Which sides of rectangle ABCD are perpendicular to side AB?

- A.** CD only
- B.** AD only
- C.** AD and BC
- D.** CD and AD

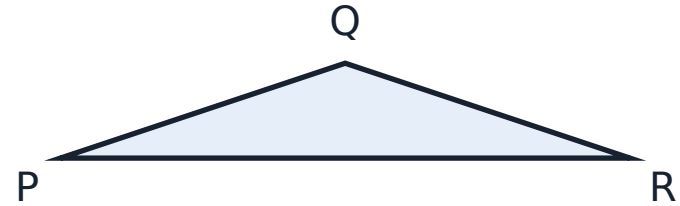


True or false?

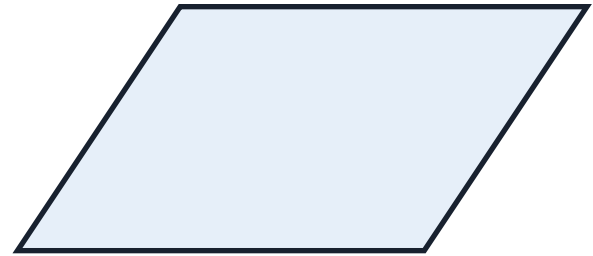
In triangle PQR, angle Q is obtuse.

A. True

B. False



How many line segments make up the sides of the figure? How many of its angles are acute?



Segments: _____

Acute angles: _____

SET 73 • 4.G.A.1

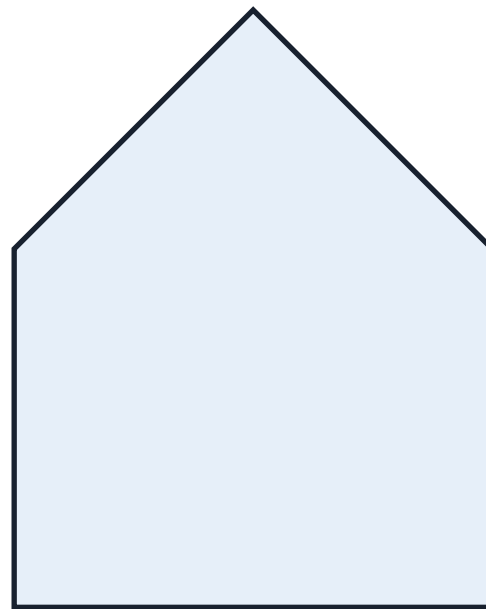
BACKWARD 1

Count sides and angles of shapes • Grade 2 • Geometry • 2.G.A.1

How many sides and how many angles does the shape have?

Sides: _____

Angles: _____



True or false?

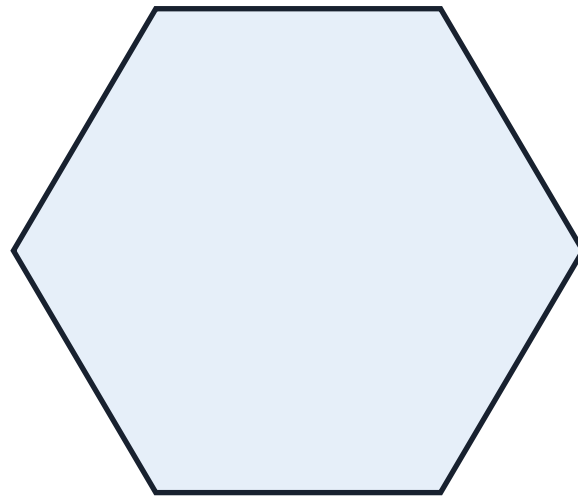
A quadrilateral has 4 sides and 4 angles.

A. True

B. False

What is the shape called?

- A.** Hexagon
- B.** Quadrilateral
- C.** Pentagon
- D.** Triangle



How many corners does a triangle have?

Corners: _____

True or false?

A pentagon has 4 angles.

A. True

B. False

SET 73 • 4.G.A.1

BACKWARD 2

Find square corners in shapes • Grade 3 • Geometry • 3.G.A.1

How many square corners
does the shape have?

Square corners: _____



True or false?

A square has 4 square corners.

A. True

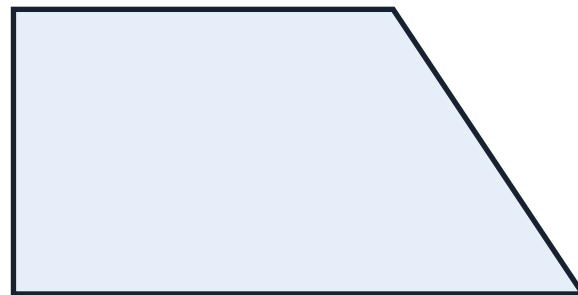
B. False

Which shape can have exactly one square corner?

- A.** Circle
- B.** Rectangle
- C.** Square
- D.** Triangle

How many square corners
does the shape have?

Square corners: _____



True or false?

A triangle can have 2 square corners.

A. True

B. False

SET 73 • 4.G.A.1

FORWARD 1

Use the attributes of a category to find sides and angles in figures • Grade 5 •
Geometry • 5.G.B.3

True or false?

Every rectangle has 4 right angles. A square is a rectangle, so every square has 4 right angles.

A. True

B. False

A figure is a rhombus.

How many pairs of parallel sides must it have?

Pairs: _____

A parallelogram has one right angle.
How many right angles must it have?

- A.** 4
- B.** 2
- C.** 3
- D.** 1

ABCD is a rectangle.

Which side is parallel to side AB?

Side: _____



True or false?

Every trapezoid has a pair of perpendicular sides.

A. True

B. False

SET 73 • 4.G.A.1

FORWARD 2

Identify parallel and perpendicular sides of polygons in the coordinate plane •

Grade 6 • Geometry • 6.G.A.3

True or false?

In the polygon with vertices $(0, 0)$, $(6, 0)$, $(6, 4)$, and $(0, 4)$, the side from $(0, 0)$ to $(6, 0)$ is parallel to the side from $(0, 4)$ to $(6, 4)$.

A. True

B. False

A triangle has vertices $(1, 1)$, $(1, 6)$, and $(5, 1)$.

At which vertex is the right angle?

Vertex: _____

A quadrilateral has vertices $(-3, 0)$, $(3, 0)$, $(1, 4)$, and $(-1, 4)$.

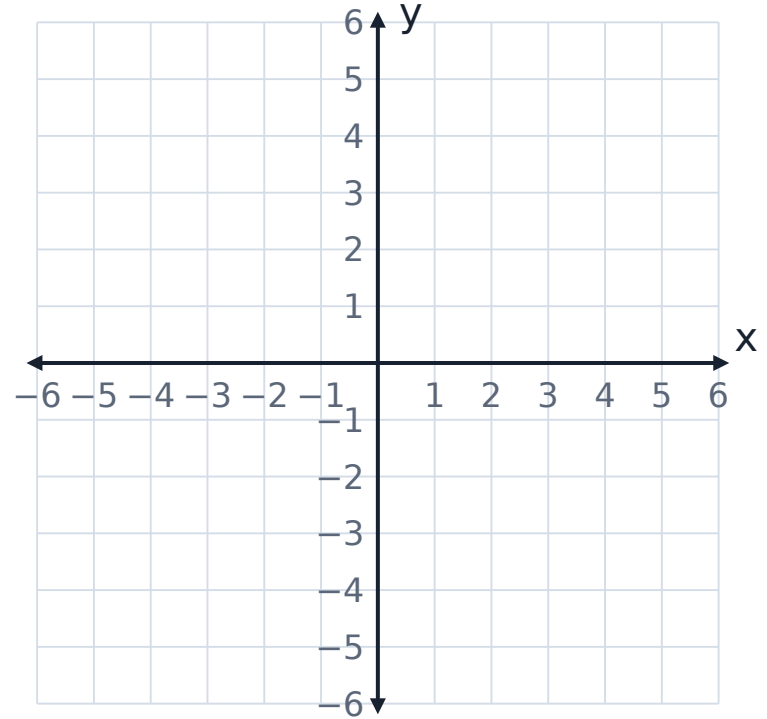
Which two sides are parallel?

- A.** No two sides are parallel **B.** From $(-3, 0)$ to $(3, 0)$ and from $(-1, 4)$ to $(1, 4)$
- C.** From $(3, 0)$ to $(1, 4)$ and from $(-1, 4)$ to $(-3, 0)$ **D.** Every pair of sides is parallel

Plot $(-4, -1)$, $(2, -1)$, $(2, 3)$, and $(-4, 3)$, and connect them in order.

How many right angles does the polygon have?

Right angles: _____



True or false?

The triangle with vertices $(0, 0)$, $(4, 0)$, and $(2, 3)$ has a right angle.

A. True

B. False

Classify two-dimensional figures by parallel or perpendicular sides and by angles

MAIN	Grade 4 main question	4.G.A.2
BACKWARD 1	Quadrilaterals as a category	3.G.A.1
BACKWARD 2	Triangles, quadrilaterals, pentagons, and hexagons	2.G.A.1
FORWARD 1	Classify quadrilaterals in a hierarchy	5.G.B.4
FORWARD 2	Find the area of parallelograms, rhombuses, and trapezoids	6.G.A.1

SET 61 • 4.G.A.2

MAIN

Grade 4 main question • Grade 4 • Geometry • 4.G.A.2

Which shape has two pairs of parallel sides and no right angles?

- A.** A square
- B.** A rhombus that is not a square
- C.** A rectangle
- D.** A right triangle

A quadrilateral has exactly one pair of parallel sides.

What is it called?

Name: _____

True or false?

A rectangle has two pairs of parallel sides and four right angles.

A. True

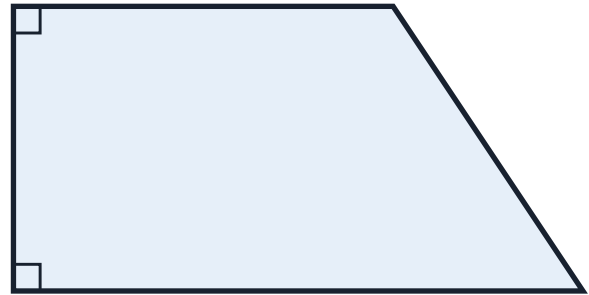
B. False

Which shape always has perpendicular sides?

- A.** Rhombus
- B.** Parallelogram
- C.** Square
- D.** Trapezoid

The shape has one pair of parallel sides and two right angles.

What kind of quadrilateral is it?



Name: _____

SET 61 • 4.G.A.2

BACKWARD 1

Quadrilaterals as a category • Grade 3 • Geometry • 3.G.A.1

True or false?

Squares and rectangles are both quadrilaterals.

- A.** True
- B.** False

What is a closed shape with 4 straight sides called?

Name: _____

Which shape is NOT a quadrilateral?

- A.** Square
- B.** Rectangle
- C.** Pentagon
- D.** Rhombus

True or false?

All quadrilaterals are squares.

A. True

B. False

Name two different quadrilaterals.

Shapes: _____

SET 61 • 4.G.A.2

BACKWARD 2

Triangles, quadrilaterals, pentagons, and hexagons • Grade 2 • Geometry •
2.G.A.1

What is a shape with 5 sides called?

Name: _____

True or false?

A hexagon has 6 sides.

A. True

B. False

Which shape has 4 sides?

A. Quadrilateral

B. Triangle

C. Pentagon

D. Hexagon

How many sides does a triangle have?

Sides: _____

True or false?

A pentagon has 8 sides.

A. True

B. False

SET 61 • 4.G.A.2

FORWARD 1

Classify quadrilaterals in a hierarchy • Grade 5 • Geometry • 5.G.B.4

Which list goes from the most general category to the most specific?

- A.** Parallelogram, quadrilateral, square, rhombus
- B.** Quadrilateral, parallelogram, rhombus, square
- C.** Rhombus, square, quadrilateral, parallelogram
- D.** Square, rhombus, parallelogram, quadrilateral

True or false?

Every square is a rhombus.

A. True

B. False

Which of these categories does every rectangle belong to?

quadrilateral, parallelogram, rhombus, square

Categories: _____

True or false?

Every rhombus is a rectangle.

A. True

B. False

A quadrilateral has 4 sides of equal length and no right angles.

What is its most specific name?

A. Rhombus

B. Quadrilateral

C. Square

D. Rectangle

SET 61 • 4.G.A.2

FORWARD 2

Find the area of parallelograms, rhombuses, and trapezoids • Grade 6 •
Geometry • 6.G.A.1

A parallelogram has a base of 9 cm and a height of 4 cm.

What is its area?

Area: _____

True or false?

A rhombus with a base of 6 m and a height of 5 m has an area of 30 square meters.

- A. True
- B. False

A trapezoid has bases of 4 ft and 8 ft and a height of 5 ft.

What is its area?

A. 40 square feet

B. 17 square feet

C. 30 square feet

D. 60 square feet

A trapezoid has bases of 6 units and 10 units and a height of 3 units.

What is its area?

Area: _____

True or false?

A parallelogram with a base of 10 units, slanted sides of 6 units, and a height of 5 units has an area of 60 square units.

A. True

B. False

Recognize right triangles as a category, and identify them

MAIN	Grade 4 main question	4.G.A.2
BACKWARD 1	Triangles	2.G.A.1
BACKWARD 2	Square corners	3.G.A.1
FORWARD 1	Right triangles share the attributes of all triangles	5.G.B.3
FORWARD 2	Find the area of right triangles	6.G.A.1

SET 62 • 4.G.A.2

MAIN

Grade 4 main question • Grade 4 • Geometry • 4.G.A.2

A triangle has angles of 90° , 50° , and 40° .
Is it a right triangle? Explain.

Answer: _____

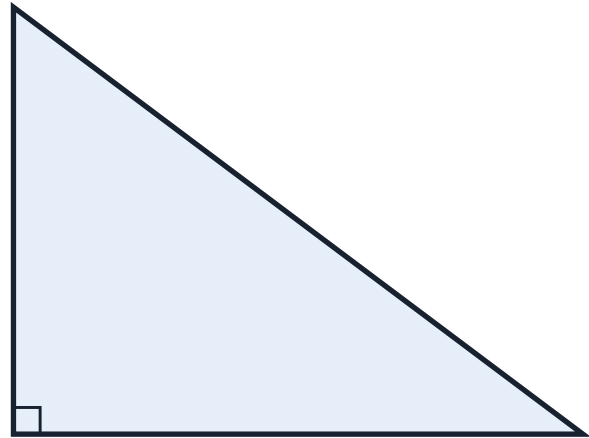
Explanation: _____

True or false?

The triangle shown is a right triangle.

A. True

B. False



Which could be the angles of a right triangle?

- A.** $70^\circ, 70^\circ, 40^\circ$
- B.** $100^\circ, 50^\circ, 30^\circ$
- C.** $90^\circ, 60^\circ, 30^\circ$
- D.** $80^\circ, 60^\circ, 40^\circ$

True or false?

A right triangle can have two right angles.

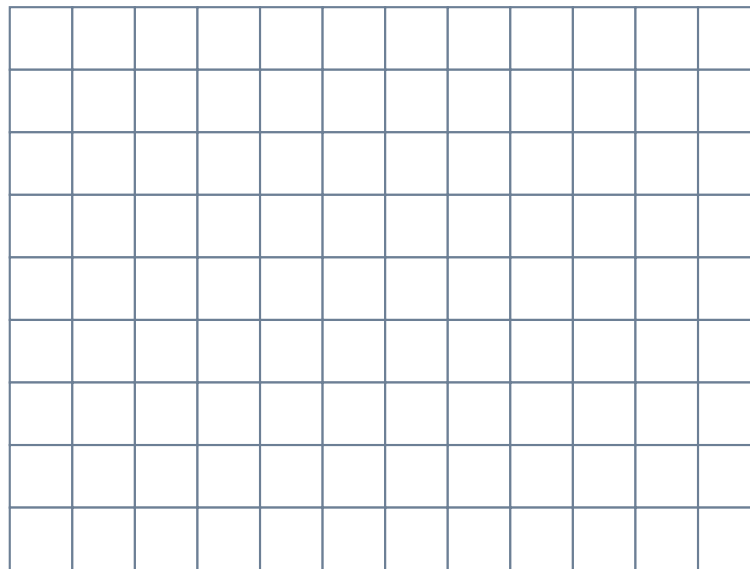
A. True

B. False

Draw a right triangle on the grid and mark its right angle.

How many right angles does your triangle have?

Right angles: _____



SET 62 • 4.G.A.2

BACKWARD 1

Triangles • Grade 2 • Geometry • 2.G.A.1

How many sides does every triangle have?

Sides: _____

True or false?

A triangle has 3 corners.

A. True

B. False

Which describes a triangle?

- A.** A closed shape with 4 straight sides
- B.** A closed shape with 3 straight sides
- C.** A round shape
- D.** A closed shape with 5 corners

How many angles does a triangle have?

Angles: _____

True or false?

All triangles are the same size.

A. True

B. False

SET 62 • 4.G.A.2

BACKWARD 2

Square corners • Grade 3 • Geometry • 3.G.A.1

True or false?

A square corner is a right angle.

A. True

B. False

Which shape always has a square corner?

- A.** Oval
- B.** Hexagon
- C.** Rectangle
- D.** Circle

How many square corners does a rectangle have?

Square corners: _____

True or false?

A triangle can never have a square corner.

A. True

B. False

Which can you use to check for a square corner: the corner of a sheet of paper or a pencil?

Answer: _____

SET 62 • 4.G.A.2

FORWARD 1

Right triangles share the attributes of all triangles • Grade 5 • Geometry •
5.G.B.3

True or false?

All triangles have 3 sides. A right triangle is a triangle, so it has 3 sides.

A. True

B. False

An isosceles right triangle belongs to which categories?

- A.** Right triangles and isosceles triangles
- B.** Only isosceles triangles
- C.** Only right triangles
- D.** Equilateral triangles

Can a right triangle be scalene? Write yes or no.

Answer: _____

True or false?

An equilateral triangle can be a right triangle.

- A. True
- B. False

A right triangle has sides of 3, 4, and 5 units.
Is it isosceles, equilateral, or scalene?

Answer: _____

SET 62 • 4.G.A.2

FORWARD 2

Find the area of right triangles • Grade 6 • Geometry • 6.G.A.1

A right triangle has legs that are 8 m and 5 m long.

What is its area?

Area: _____

True or false?

Two copies of the same right triangle can be put together to make a rectangle.

- A.** True
- B.** False

A right triangle has legs that are 12 in. and 6 in. long.

What is its area?

- A.** 9 square inches
- B.** 36 square inches
- C.** 18 square inches
- D.** 72 square inches

A 10-by-4 rectangle is cut along a diagonal into two right triangles.

What is the area of each triangle?

Area: _____

True or false?

A right triangle with legs of 7 and 4 units has an area of 28 square units.

A. True

B. False

Recognize lines of symmetry and line-symmetric figures

MAIN	Grade 4 main question	4.G.A.3
BACKWARD 1	Split circles and rectangles into equal shares	2.G.A.3
BACKWARD 2	Halves and fourths	1.G.A.3
FORWARD 1	Graph points that match across a line of symmetry	5.G.A.2
FORWARD 2	Reflect points across the axes	6.NS.C.6.b

SET 63 • 4.G.A.3

MAIN

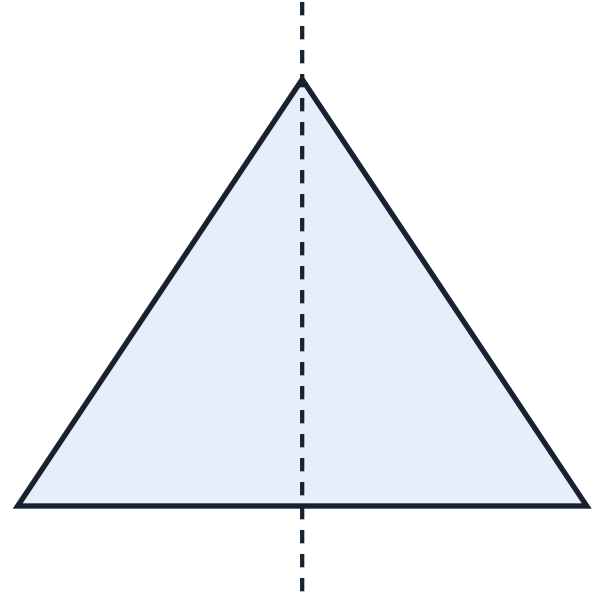
Grade 4 main question • Grade 4 • Geometry • 4.G.A.3

True or false?

The dashed line is a line of symmetry of the triangle.

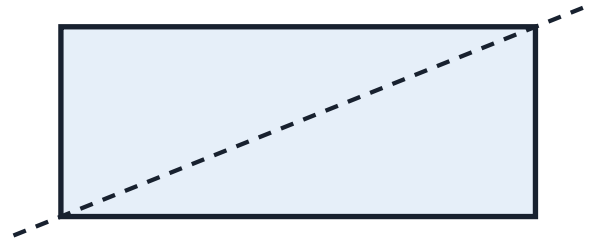
A. True

B. False



True or false?

The dashed line is a line of symmetry of the rectangle.



A. True

B. False

How many lines of symmetry does a square have?

A. 1

B. 4

C. 0

D. 2

How many lines of symmetry does a rectangle that is not a square have?

Lines: _____

Which capital letter has a vertical line of symmetry?

A. R

B. J

C. A

D. F

SET 63 • 4.G.A.3

BACKWARD 1

Split circles and rectangles into equal shares • Grade 2 • Geometry • 2.G.A.3

True or false?

A rectangle cut into 2 equal shares is cut into halves.

A. True

B. False

A circle is cut into 4 equal shares.

What is one share called?

Name: _____

A rectangle is cut into 3 equal shares.

What are the shares called?

- A.** Fifths
- B.** Halves
- C.** Thirds
- D.** Fourths

How many halves make a whole?

Halves: _____

True or false?

Equal shares of the same whole must have the same shape.

- A.** True
- B.** False

SET 63 • 4.G.A.3

BACKWARD 2

Halves and fourths • Grade 1 • Geometry • 1.G.A.3

A square is cut into 2 equal parts.

What is each part called?

Name: _____

True or false?

Cutting a circle into 4 equal parts makes smaller parts than cutting it into 2 equal parts.

A. True

B. False

How many fourths make a whole?

A. 2

B. 8

C. 4

D. 1

A pizza is cut into 4 equal parts.

What is one part called?

Name: _____

True or false?

Halves are smaller than fourths of the same whole.

A. True

B. False

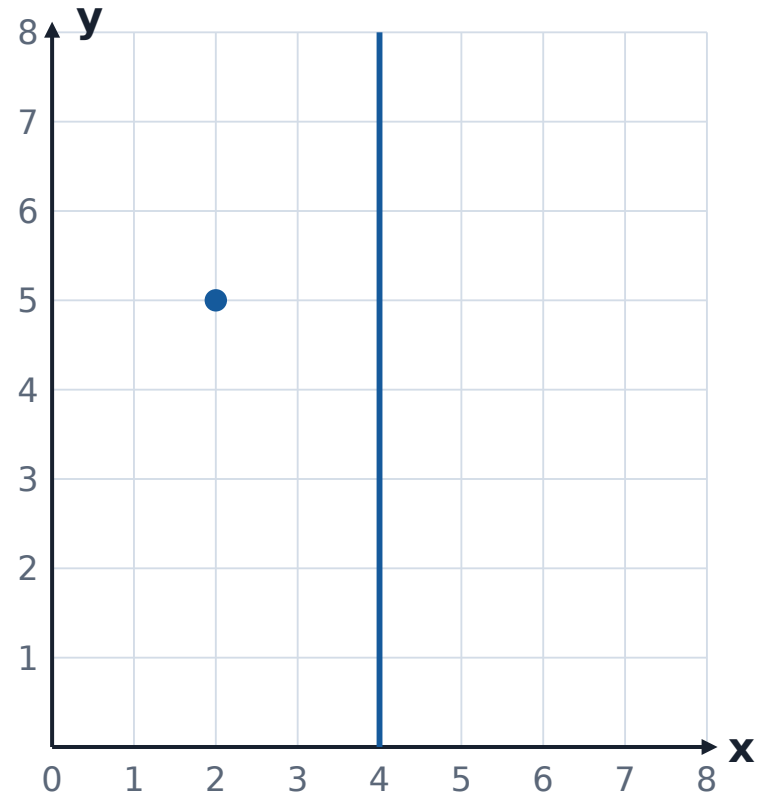
SET 63 • 4.G.A.3

FORWARD 1

Graph points that match across a line of symmetry • Grade 5 • Geometry •
5.G.A.2

The line through $(4, 0)$ and $(4, 8)$ is a line of symmetry.

Plot the point that matches $(2, 5)$ on the other side of the line.



True or false?

Across the line through $(3, 0)$ and $(3, 5)$, the point $(1, 2)$ matches the point $(5, 2)$.

A. True

B. False

Across the line through $(5, 0)$ and $(5, 8)$, which point matches $(7, 4)$?

A. $(5, 4)$

B. $(2, 4)$

C. $(7, 6)$

D. $(3, 4)$

Across the line through $(4, 0)$ and $(4, 8)$, which point matches $(1, 6)$?

Point: _____

True or false?

Across the line through $(2, 0)$ and $(2, 5)$, the point $(0, 3)$ matches the point $(3, 3)$.

A. True

B. False

SET 63 • 4.G.A.3

FORWARD 2

Reflect points across the axes • Grade 6 • The Number System • 6.NS.C.6.b

Reflect $(3, 4)$ across the y -axis.

What are the coordinates of the image?

Image: _____

True or false?

Reflecting $(2, -5)$ across the x-axis gives $(2, 5)$.

- A. True
- B. False

Which point is the reflection of $(-4, 1)$ across the x-axis?

- A.** $(4, 1)$
- B.** $(-4, -1)$
- C.** $(1, -4)$
- D.** $(4, -1)$

Reflect $(-6, -2)$ across the y -axis.

What are the coordinates of the image?

Image: _____

True or false?

Reflecting $(5, 0)$ across the x-axis gives $(-5, 0)$.

A. True

B. False

Draw lines of symmetry

MAIN	Grade 4 main question	4.G.A.3
BACKWARD 1	Split circles and rectangles into equal shares	2.G.A.3
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SET 64 • 4.G.A.3

MAIN

Grade 4 main question • Grade 4 • Geometry • 4.G.A.3

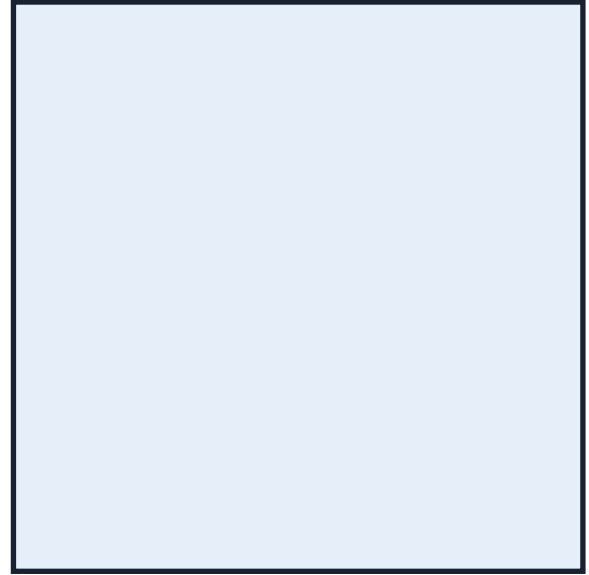
Draw all the lines of symmetry of the rectangle.



Draw all the lines of
symmetry of the square.

How many are there?

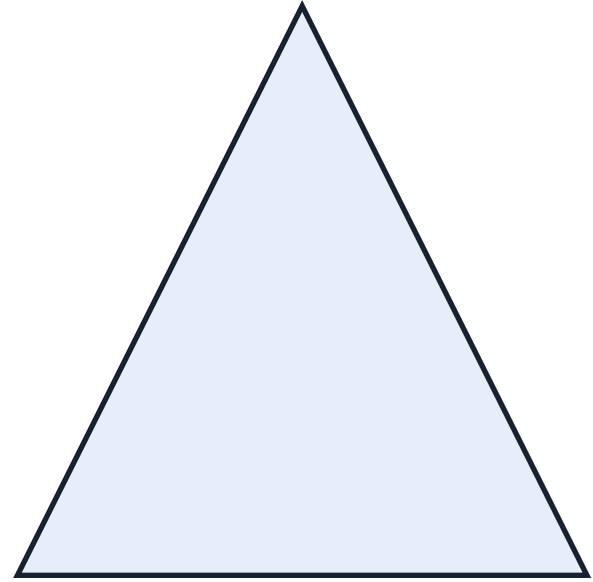
Lines: _____



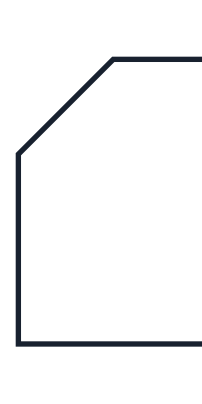
Draw the line of symmetry of the triangle. The triangle has exactly two sides of equal length; its base has a different length.

How many lines of symmetry does it have?

Lines: _____



Complete the figure so that the dashed line is a line of symmetry.



A regular hexagon has 6 equal sides and 6 equal angles.
How many lines of symmetry does it have?

Lines: _____

SET 64 • 4.G.A.3

BACKWARD 1

Split circles and rectangles into equal shares • Grade 2 • Geometry • 2.G.A.3

A square is cut into 4 equal shares.

How many shares make the whole?

Shares: _____

True or false?

A circle can be cut into 3 equal shares called thirds.

A. True

B. False

Which one shows halves?

- A.** A circle cut into 4 equal parts
- B.** A rectangle cut into 2 unequal parts
- C.** A rectangle cut into 2 equal parts
- D.** A circle cut into 3 equal parts

A rectangle is cut into 3 equal shares.

What is one share called?

Name: _____

True or false?

A circle cut into 2 parts of different sizes shows halves.

A. True

B. False

SET 64 • 4.G.A.3

BACKWARD 2

Put shapes together to make new shapes • Grade 1 • Geometry • 1.G.A.2

True or false?

Two matching right triangles can be put together to make a rectangle.

- A.** True
- B.** False

Which two shapes can be put together to make a square?

- A.** Two matching rectangles
- B.** Two circles
- C.** Three circles
- D.** A circle and a triangle

How many squares do you need to make a rectangle that is 2 squares long and 1 square wide?

Squares: _____

True or false?

Two half-circles can be put together to make a square.

A. True

B. False

Two matching trapezoids are put together to make a shape with 6 sides.

What is the shape called?

Name: _____

SET 64 • 4.G.A.3

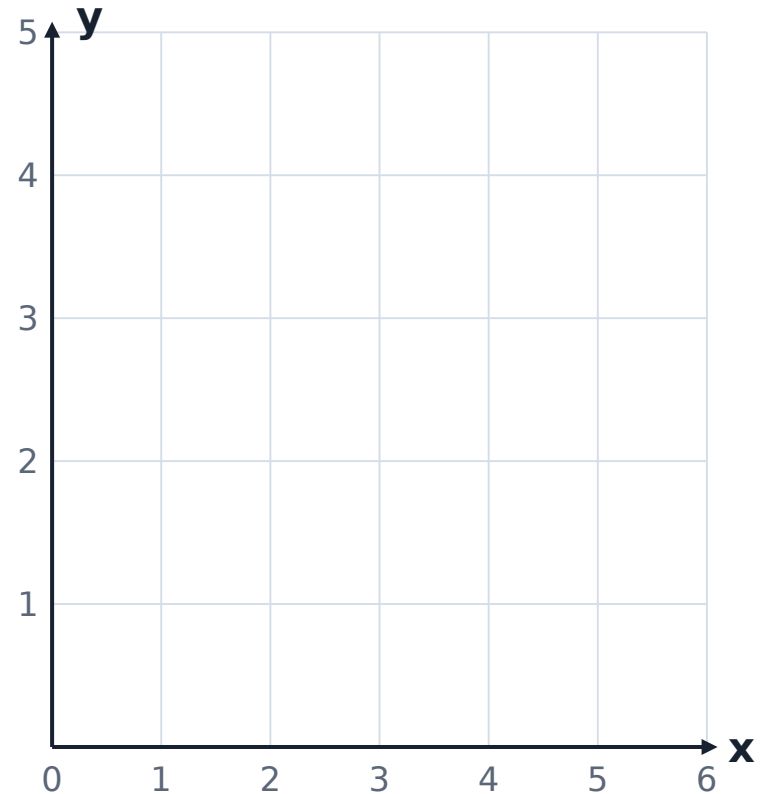
FORWARD 1

Graph points to make and check symmetric figures • Grade 5 • Geometry •
5.G.A.2

Plot $(1, 1)$, $(3, 4)$, and $(5, 1)$ and connect them. Then draw the line of symmetry.

Where does the line of symmetry cross the x-axis?

Crosses at: _____



True or false?

The rectangle with corners $(1, 1)$, $(5, 1)$, $(5, 3)$, and $(1, 3)$ is symmetric about the line through $(3, 0)$ and $(3, 4)$.

A. True

B. False

A shape has a line of symmetry through $(4, 0)$ and $(4, 6)$. The point $(2, 3)$ is on the shape.

Which point must also be on it?

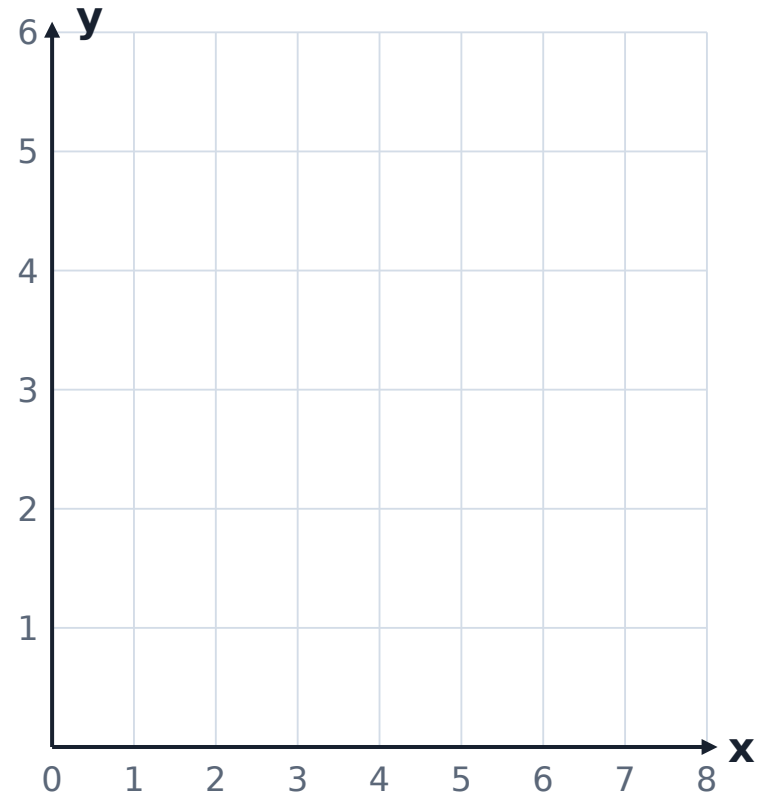
A. $(2, 5)$

B. $(4, 3)$

C. $(8, 3)$

D. $(6, 3)$

Plot $(2, 2)$, $(2, 5)$, and $(6, 2)$.
Then plot one more point so
that the four points are the
corners of a rectangle.



True or false?

The points $(1, 4)$ and $(7, 4)$ match across the line through $(3, 0)$ and $(3, 6)$.

A. True

B. False

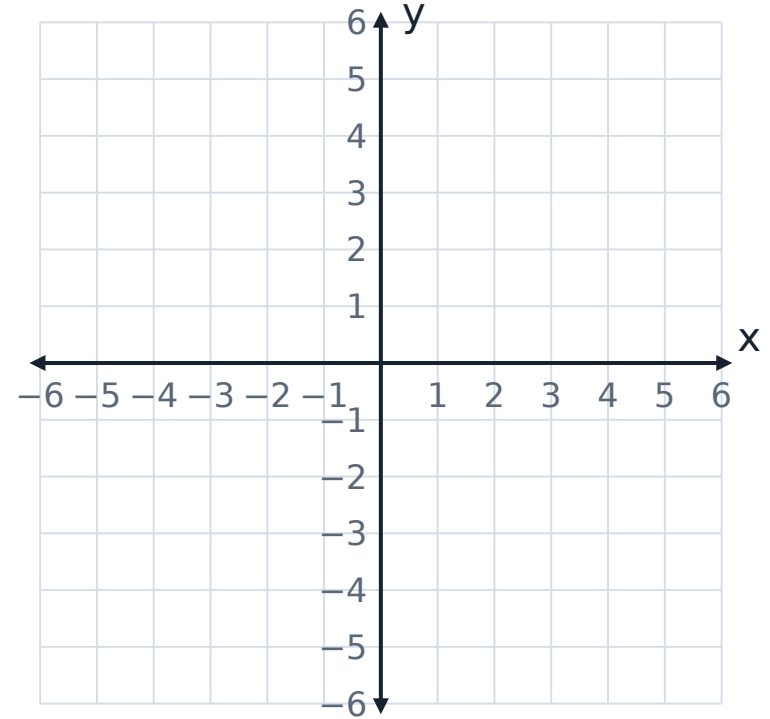
SET 64 • 4.G.A.3

FORWARD 2

Draw polygons in the coordinate plane from their vertices and find side lengths

• Grade 6 • Geometry • 6.G.A.3

Plot $(2, 3)$, $(5, 3)$, $(5, -1)$, and $(2, -1)$. Connect them in order to draw the polygon.



True or false?

The polygon with vertices $(-3, 2)$, $(3, 2)$, $(3, -2)$, and $(-3, -2)$ has a side that is 6 units long.

A. True

B. False

A square has vertices $(-2, -2)$, $(2, -2)$, and $(2, 2)$.

What is the fourth vertex?

A. $(-2, 2)$

B. $(2, -4)$

C. $(-2, 4)$

D. $(0, 2)$

A polygon has vertices $(1, 4)$, $(4, 4)$, and $(4, -2)$.

What is the length of the side from $(4, 4)$ to $(4, -2)$?

Length: _____

True or false?

The segment from $(-4, 1)$ to $(4, 1)$ is 4 units long.

- A. True
- B. False